

CS 439 25F Project

Economic Input-Output Model



Group 23

Ryan Vukicevic (rcv35)

Lawrence Ho (lh811)

Paolo Gervasoni (pg681)

Project Description

- We are looking to analyze the difference between **25 different national economies** and their structures given **Input-Output models** and potentially **forecast certain effects of the economies** .
- We will compute the **Leontief matrices** for each country using their respective input–output tables. This will allow us to analyze the structure of **inter-industry relationships** within each economy to identify key sectors.

$$\bigcirc \mathbf{x} = \mathbf{Ax} + \mathbf{y} \Rightarrow \mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{y}$$

Dataset Description and Sample Snapshots

- As illustrated by screenshots, the original dataset was an **Excel spreadsheet of 172 MB**, the **CSV conversion** for faster data loading is still above the threshold for full bonus points at **123 MB** in file explorer, and 120 MB according to Pandas
- Screenshots give **Dataset Size** , **Record Size** , **#Records** , **Entry Size**

 Ir_wiod_wiot_wide		11/4/2025 7:54 PM	Comma Separated...	123,182 KB
 Ir_wiod_wiot_wide		11/4/2025 1:38 PM	Microsoft Excel W...	172,506 KB

```
df.info(memory_usage='deep')
```

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RangeIndex: 21816 entries, 0 to 21815  
Columns: 709 entries, Year to xTOT_xGO  
dtypes: float64(706), int64(1), object(2)  
memory usage: 120.4 MB
```

Dataset Description and Sample Snapshots

- Dataset Name: **Initial WIOD** (World Income-Outcome Database) Project
- Link: <https://www.rug.nl/ggdc/valuechain/long-run-wiod>
 - It covers **25 countries** , and a category for the rest of the world, from **1965-2000**
 - Interactions between **ISIC sectors** , sector-sector transactions across nations
 - The sector in each row is the seller, the columns are the buyers, the amount is in **millions USD**
 - **Components of output** in the dataset is given by
 - Expenditure by Household (C)
 - Expenditure by Government (G)
 - Gross Fixed Capital Formation (I)
 - Change in Inventory (ΔI)

Year	Sector	Agriculture, Hunting, Forestry, and Fishing	Mining and Quarrying	Food, Beverages, and Tobacco	Textiles, Textile Products, Leather, Footwear	Pulp, Paper, Printing, Publishing
1965	Agriculture, Hunting, Forestry, and Fishing	7387.654292	36.933699	22845.005310	1862.655284	717.994280
1965	Mining and Quarrying	138.926283	1249.567560	41.649197	32.518188	141.395517
1965	Food, Beverages, and Tobacco	4248.362074	13.520971	14052.767340	403.381594	237.308836

Dataset Snapshots per Table

- Each table represents **inter-sectoral transactions** for a specific country and year.
- Rows indicate **producing sectors** , while columns indicate **consuming sectors** .
- Values are expressed in **millions of USD** .
- Shown here: examples for USA, Great Britain, and China (1970), representative of the 25 country dataset.
- These data snapshots are later used to compute each country's **A (Input–Output coefficient)** and **L (Leontief inverse)** matrices.

usa.head()

	Year	Sector	Agriculture, Hunting, Forestry, and Fishing	Mining and Quarrying	Food, Beverages, and Tobacco	Textiles, Textile Products, Leather, Footwear	Pulp, Paper, Printing, Publishing	Coke, Refined Petroleum, and Nuclear Fuel
552	1965	Agriculture, Hunting, Forestry, and Fishing	7387.654292	36.933699	22845.005310	1862.655284	717.994280	0.889029
553	1965	Mining and Quarrying	138.926283	1249.567560	41.649197	32.518188	141.395517	7584.679520
554	1965	Food, Beverages, and Tobacco	4248.362074	13.520971	14052.767340	403.381594	237.308836	78.766105

china.head()

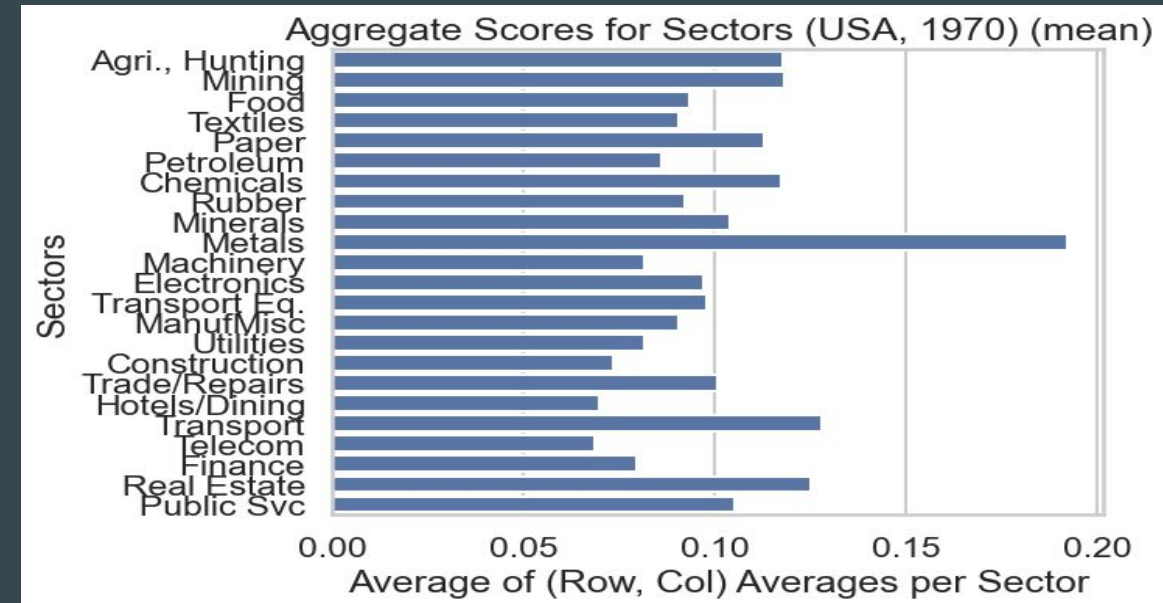
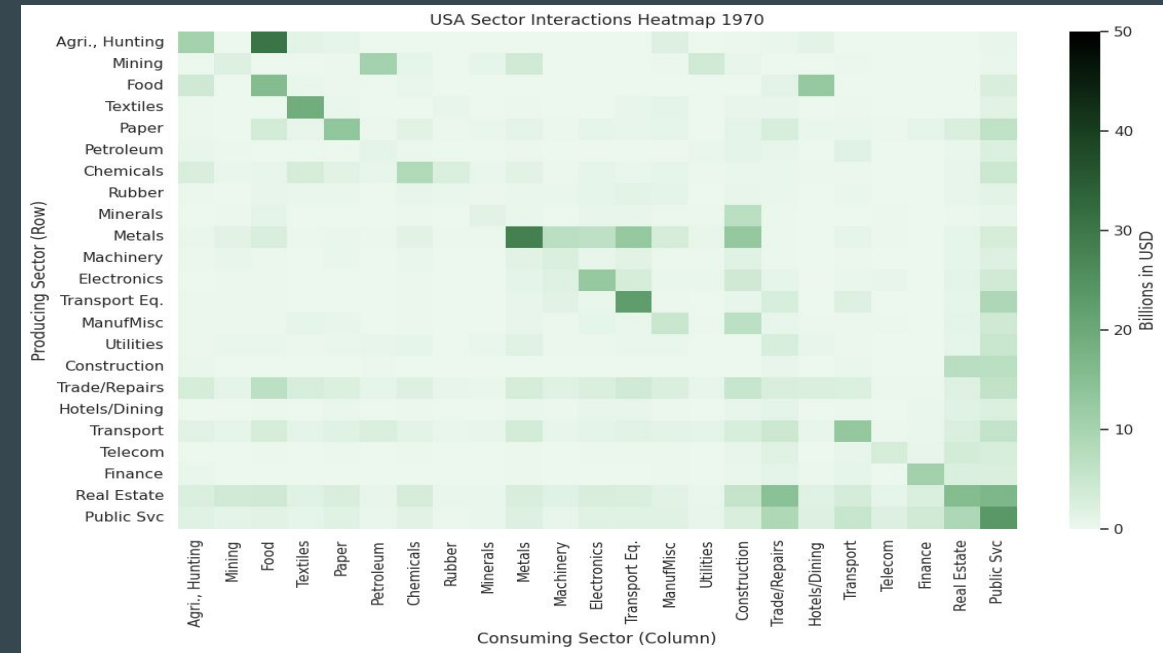
	Year	Sector	Agriculture, Hunting, Forestry, and Fishing	Mining and Quarrying	Food, Beverages, and Tobacco	Textiles, Textile Products, Leather, Footwear	Pulp, Paper, Printing, Publishing
115	1965	Agriculture, Hunting, Forestry, and Fishing	6079.321491	14.815457	3376.827067	1732.057642	278.344066
116	1965	Mining and Quarrying	15.656909	67.419013	9.892573	23.079870	13.488278
117	1965	Food, Beverages, and Tobacco	78.088996	3.394234	118.690418	24.325732	2.970446

uk.head()

	Year	Sector	Agriculture, Hunting, Forestry, and Fishing	Mining and Quarrying	Food, Beverages, and Tobacco	Textiles, Textile Products, Leather, Footwear	Pulp, Paper, Printing, Publishing	Coke, Refined Petroleum, and Nuclear Fuel
253	1965	Agriculture, Hunting, Forestry, and Fishing	904.030182	0.000000	2800.701659	292.504259	20.732738	0.000000
254	1965	Mining and Quarrying	6.323743	5.814433	12.961232	12.982891	25.240564	1177.349104
255	1965	Food, Beverages, and Tobacco	1178.182786	0.000000	3386.187471	10.216403	22.758684	11.857106

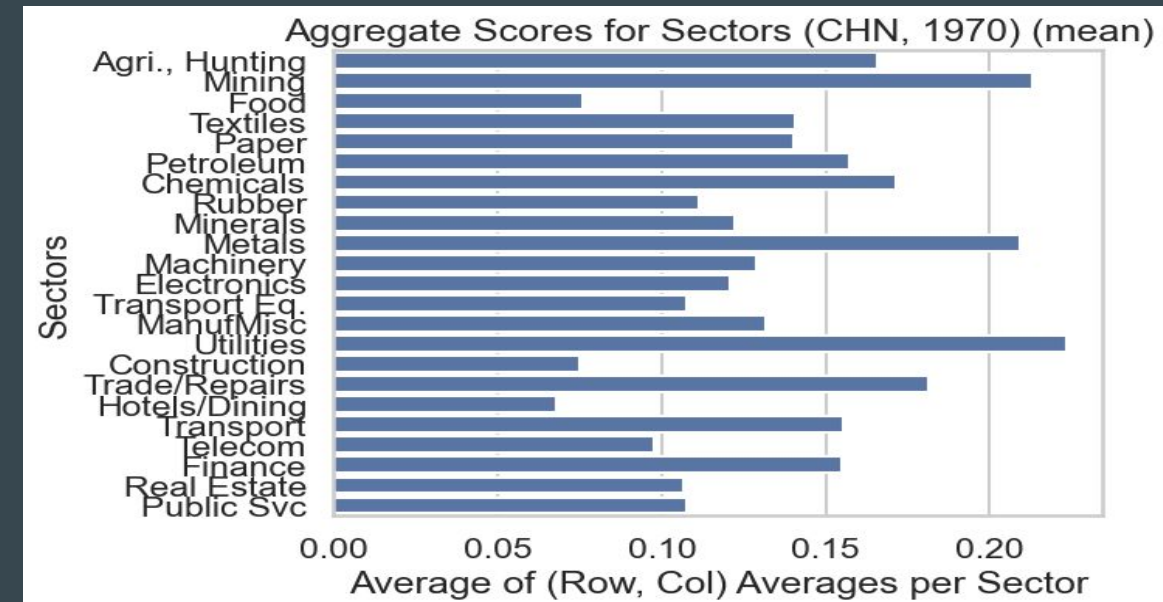
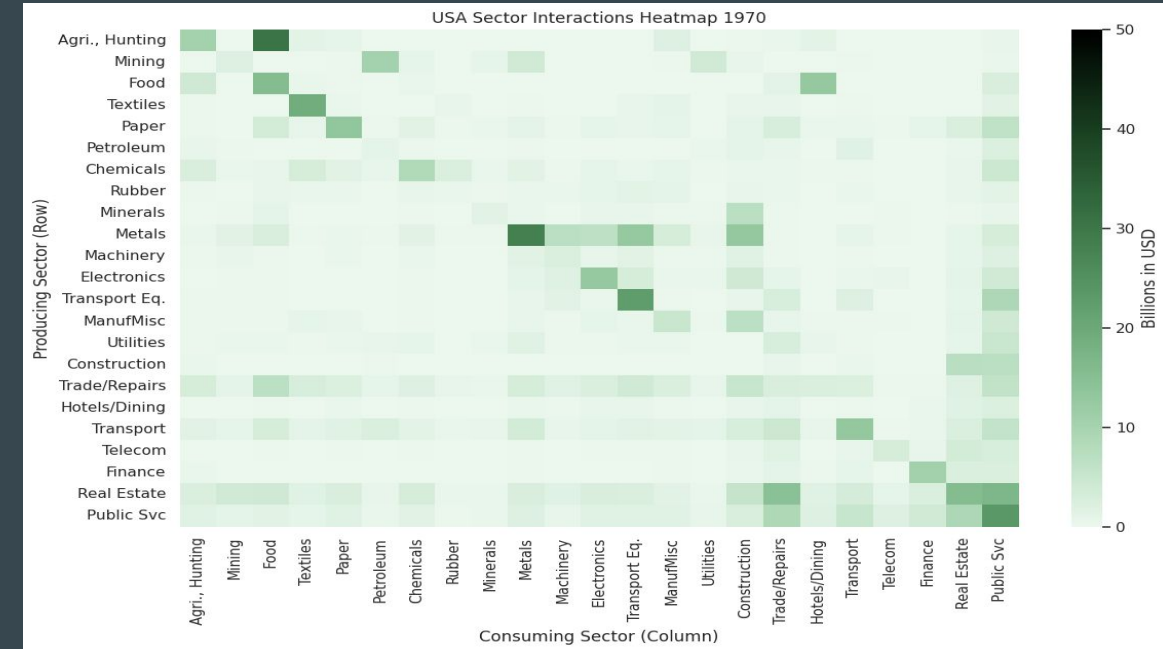
Dataset Sample Stats. Plots (USA)

- **USA 1970** Input–Output sector interactions
- **Heatmap** : Intensity of flows between producing and consuming sectors
- **Bar chart** : Average aggregate contribution per sector
- **Key sectors** : Metals, Finance, and Public Services show strongest activity
- Baseline patterns used to compute the USA's **A** and **L** matrices



Dataset Sample Stats. Plots (China)

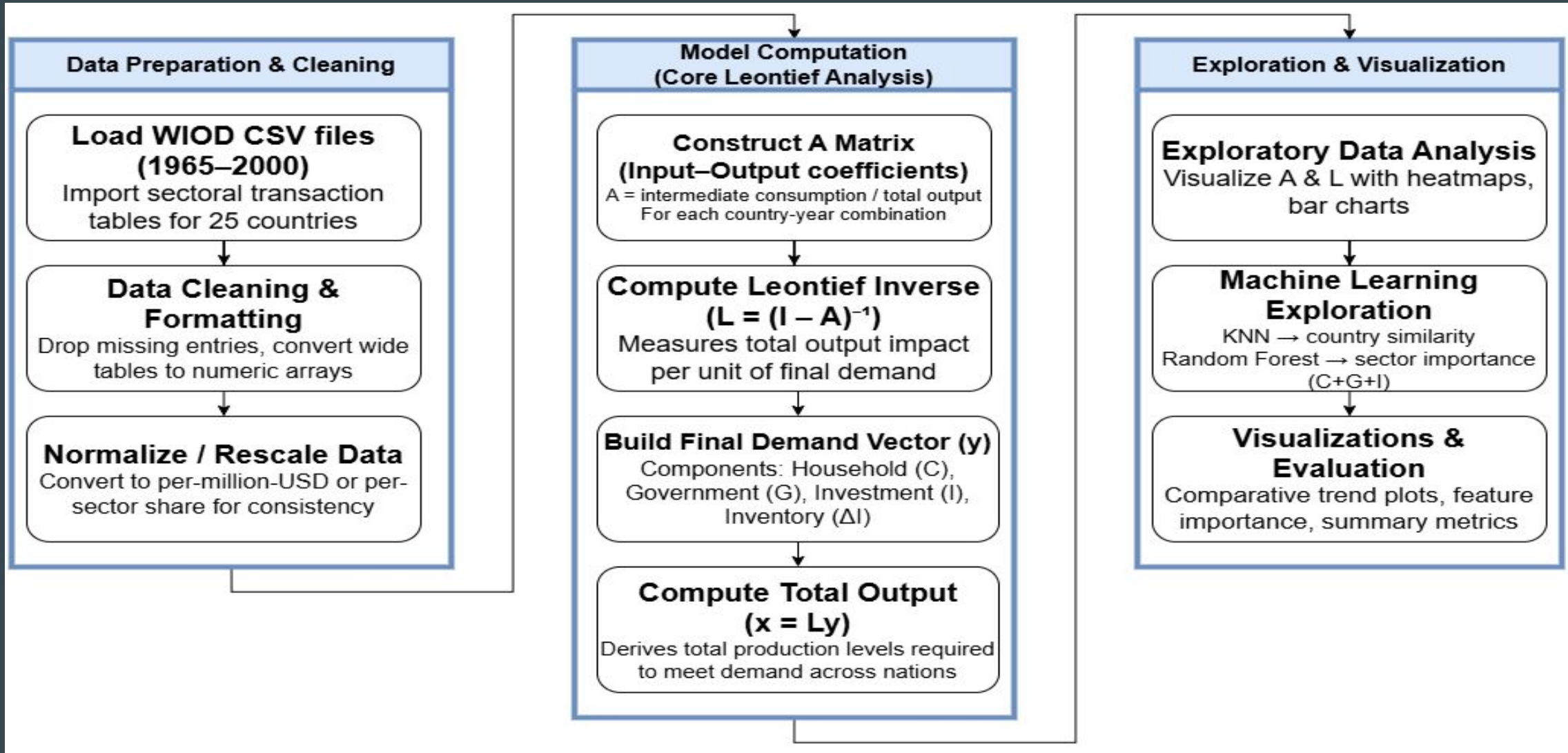
- **China 1970** Input–Output sector interactions
- **Heatmap** : Visualizes domestic inter-sector dependencies
- **Bar chart** : Highlights major sectors driving output
- **Key Sectors** : Manufacturing, Construction, and Trade
- Similar matrix structure used for global comparisons



Questions to be Addressed

- **Analyze the structural similarities/differences** between the domestic makeups of output for each economy by sector.
 - Implement **KNN Classification on L** to explore the **similarities and differences** of the countries
 - Are you able to differentiate between countries regionally given **L matrices** ?
 - Analyze potential **regional differences** ?
 - **Interpret the Leontief Matrices** to understand the **structure of economies** .
 - Using Decision trees, visualize the the splits and key decision splits of classifying regional economies.
- **Investigate the reliability of Leontief inverse matrices to classify or ('group') countries by region.**
 - Implement K-Means clustering to see if we can also precisely split the countries into the same 3 regions using the same L matrices.

Data Transformation Flowchart



Exploration Algorithms Used

- Matrix-based Input Output Analysis:
 - Implemented the **Leontief Model** ($x = Ax + y \Rightarrow x = (I - A)^{-1} y$) to quantify industry dependencies and estimate total output required to meet final demand
- Statistical Summarization:
 - Computed yearly summary statistics (mean, median, minimum, maximum) of sector-sector relationships to evaluate and compare economic growth among countries over time
- Data Visualization Algorithms:
 - Used **Seaborn** and **Matplotlib** for heatmaps, line plots, and comparative temporal trend analyses, enabling the visualization of sector interactions and national output evolution.
- Iterative Exploration Loops:
 - Developed **Python loops** to iterate across countries and years, making visual summaries and sectoral comparison matrices.

Computational Platform and Software Libraries Used

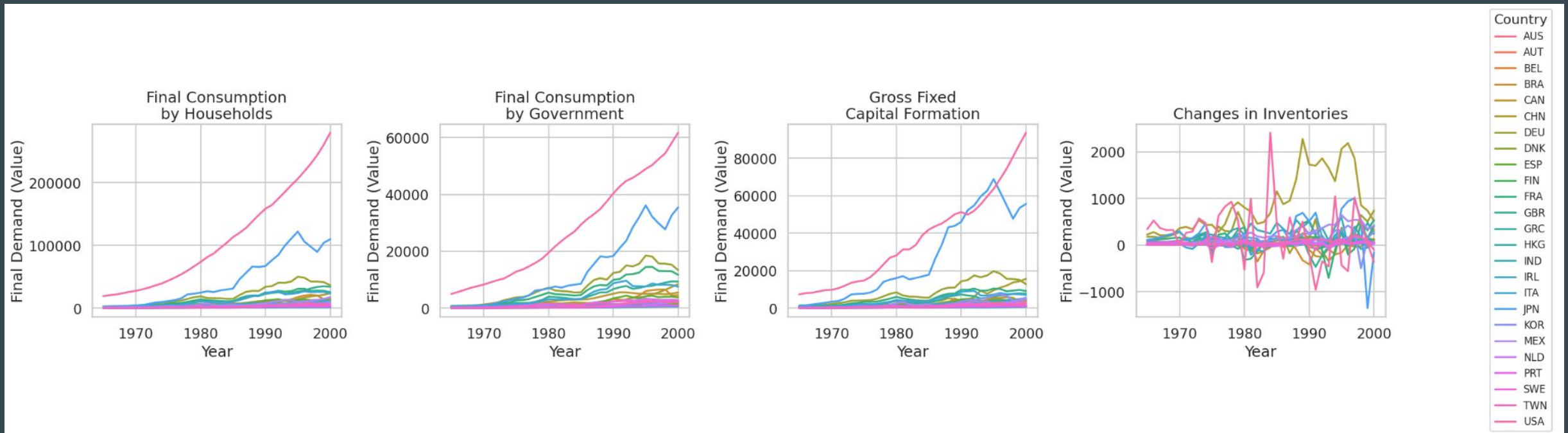
- **Development Environment :**
 - Google Colab — used for cloud-based execution of Jupyter notebooks and seamless collaboration.
- **Data Loading :**
 - Google Drive Library — accessed and read large .csv files (≈ 120 MB WIOD dataset) directly from Drive.
- **Preprocessing and Analysis:**
 - NumPy, Pandas — for data wrangling, matrix operations, and computing Input–Output coefficients.
- **Visualization :**
 - Matplotlib, Seaborn — for generating line plots, heatmaps, and sectoral trend visualizations.

Gantt Chart

Task	Person	Days
Project Setup & Data Collection	Ryan Vukicevic	1
Data Cleaning & Preprocessing	Ryan Vukicevic	2
Matrix Computation Function ($A, L, x = Ly$)	Lawrence Ho	2
More Preprocessing (Labeling Data)	Lawrence Ho	1
Visualization (Heatmaps, Plots)	Paolo Gervasoni	1.5
Evaluation & Final Presentation Prep	Paolo Gervasoni	1.5

Visualizations Samples

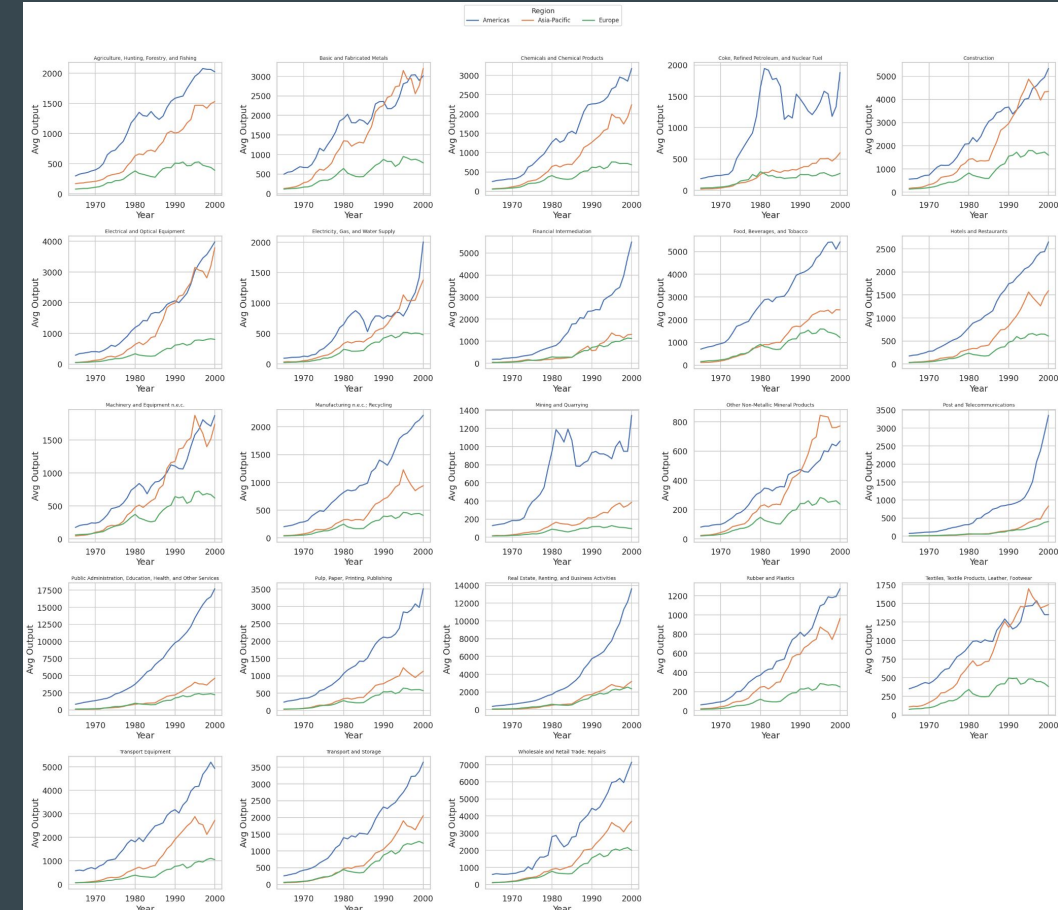
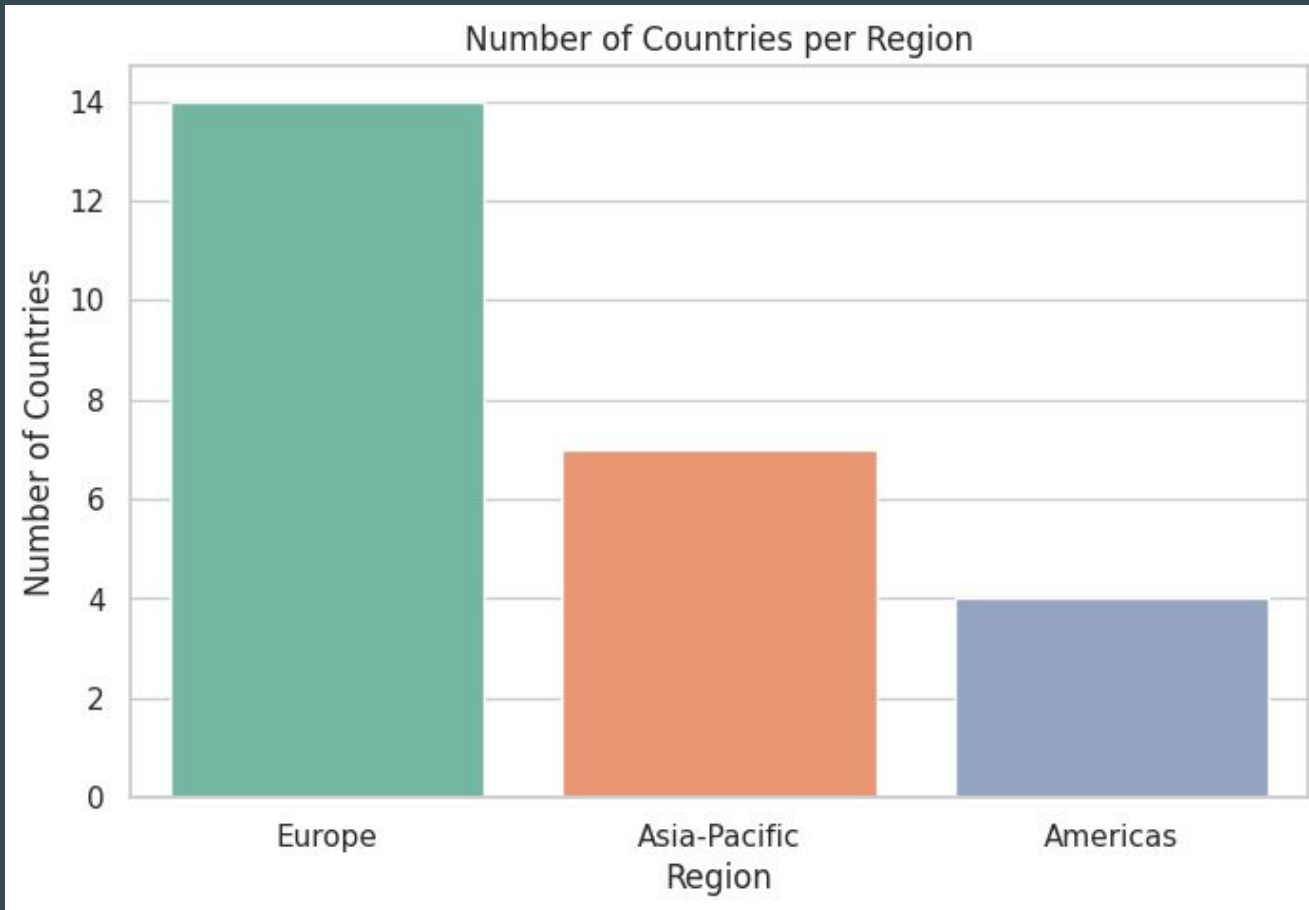
- Displays **final demand trends across 25 countries**
- Final C, G, I shows an upward trend over the years from 1965-2000. This is expected as productivity and income increases over time (without adjustment for inflation).
 - Changes in inventory tracks the volatile component of Investments in GDP.



Visualizations Samples

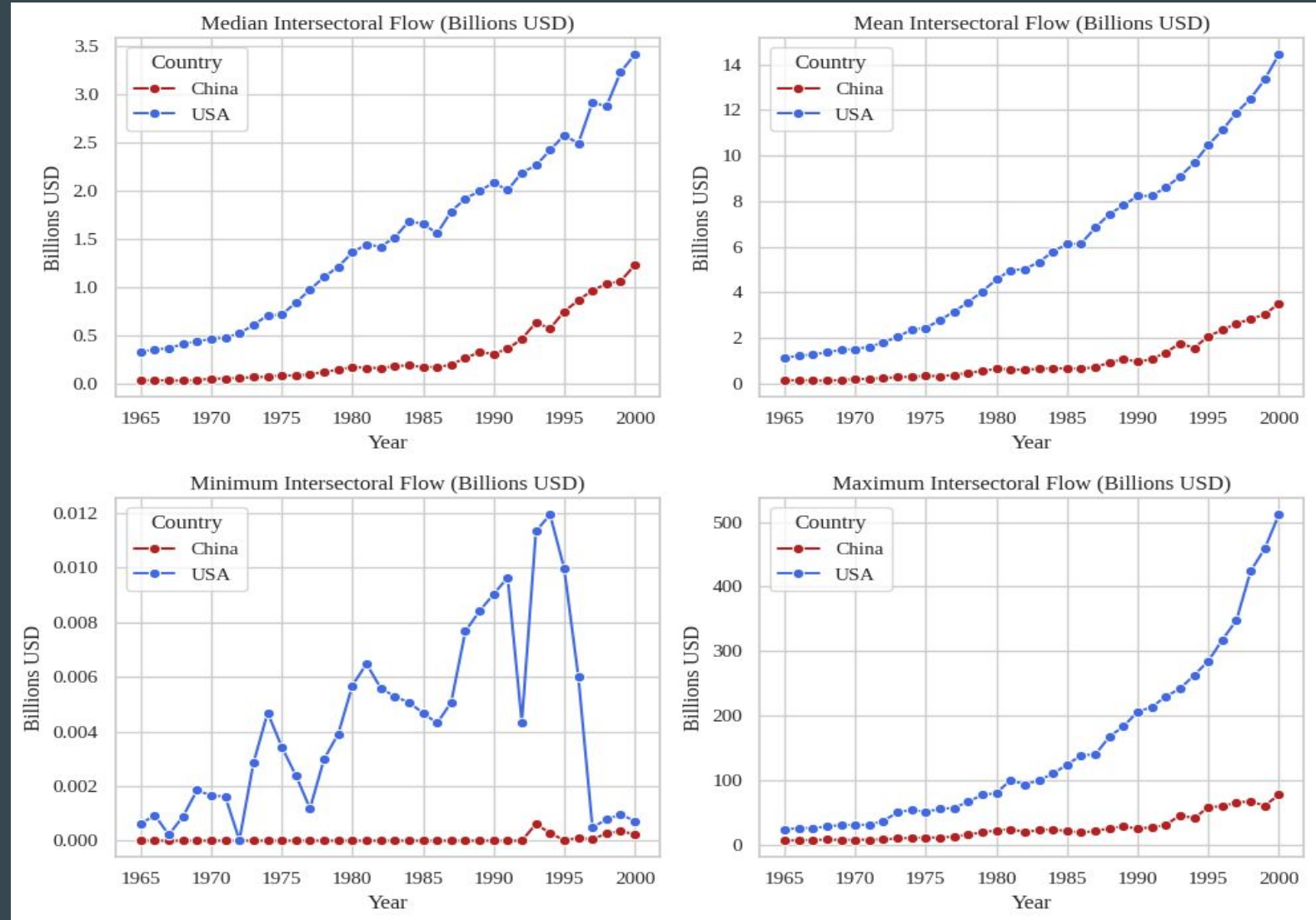
- Some visualization regarding the regions in our data

*caveat: Australia is classified as Asia-Pacific although more precise would be Oceania but they're the only country with that classification



Visualizations Samples

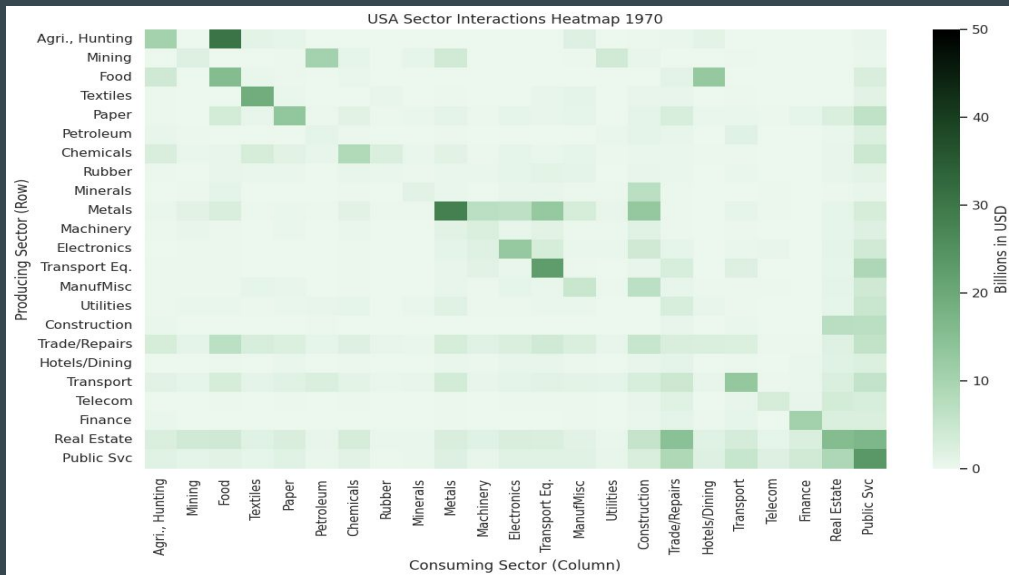
- Compares **intersectoral flow trends** across 25 economies (examples shown: USA, China).
- **Mean and median flows** steadily increase for both nations, reflecting **rising industrial connectivity**.
- The **maximum flow** in the USA shows rapid post-1980 acceleration indicating **economic scaling effects**.



Dashboard Mock Up

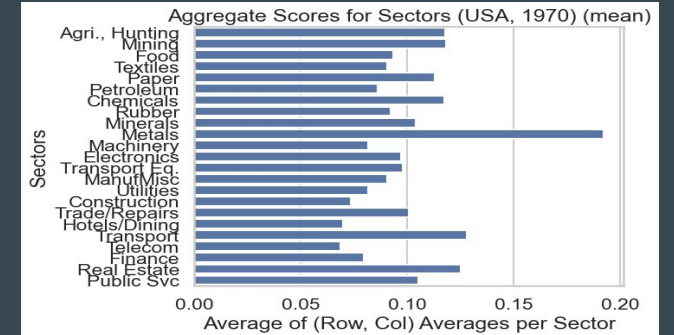
Country-Year Selector and Heatmap View

- Dropdown to select Country and Year (1965–2000)
- Displays sectoral flow heatmap (A or L matrix) for the chosen country-year



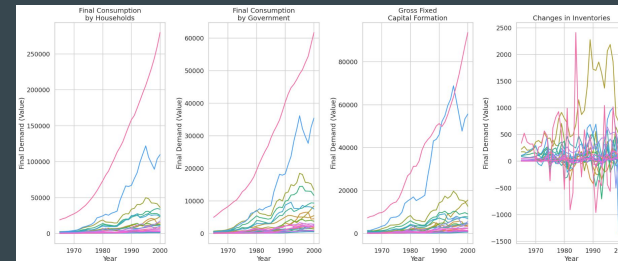
Key Sector Summary

- Shows **top 3 sectors** driving output ($C + G + I$)
- Highlights **growth rate** or **contribution %**

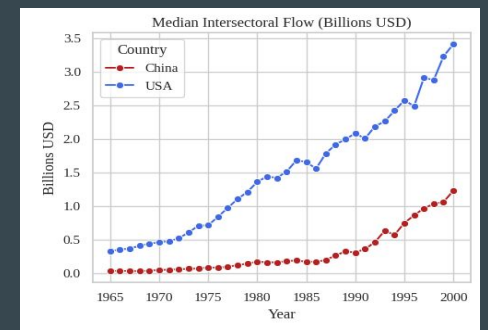


Final Output Components Trend Panel

- Displays side by side comparisons



Comparative Insights Panel



Evaluation Techniques

- **Initial Validation:**
 - Verified consistency of **A and L matrices** across **25 countries** and years (1965–2000).
 - Compared computed **total outputs** ($x = Ly$) with **WIOD reported aggregates**.
- **Model Evaluation:**
 - Used **KNN** to determine **regional industry similarity** .
 - **Random Forest** to identify **key sectors driving aggregate output** ($C + G + I$).

Interesting Findings I

KNN Results

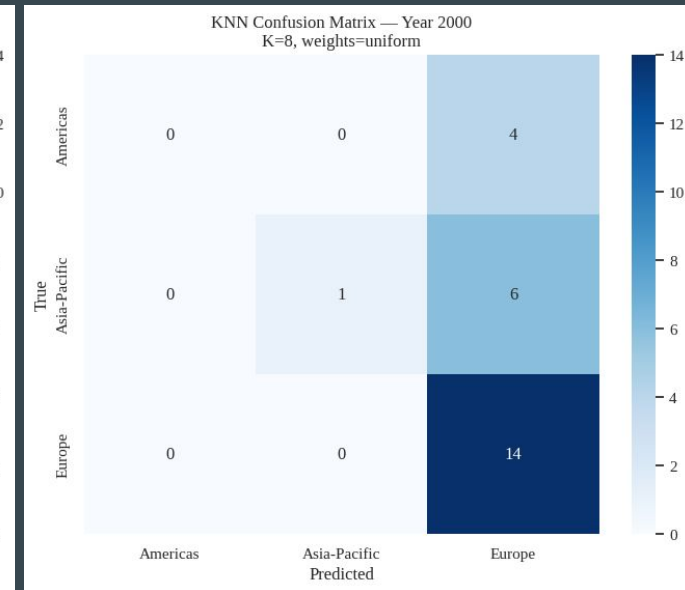
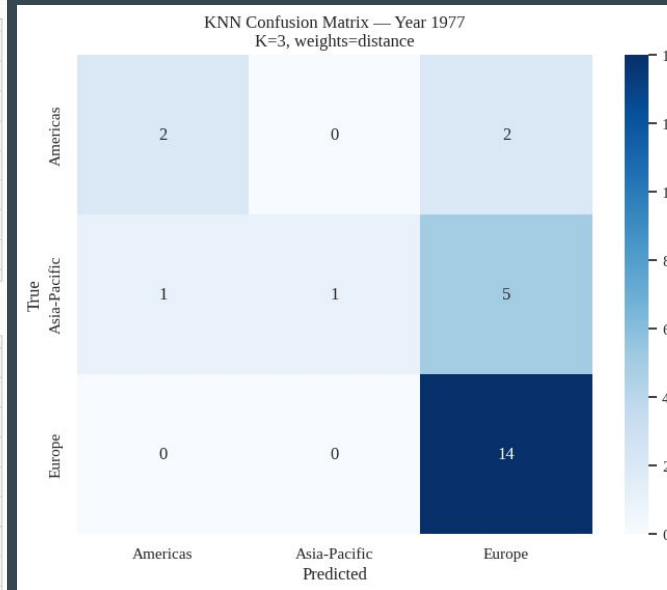
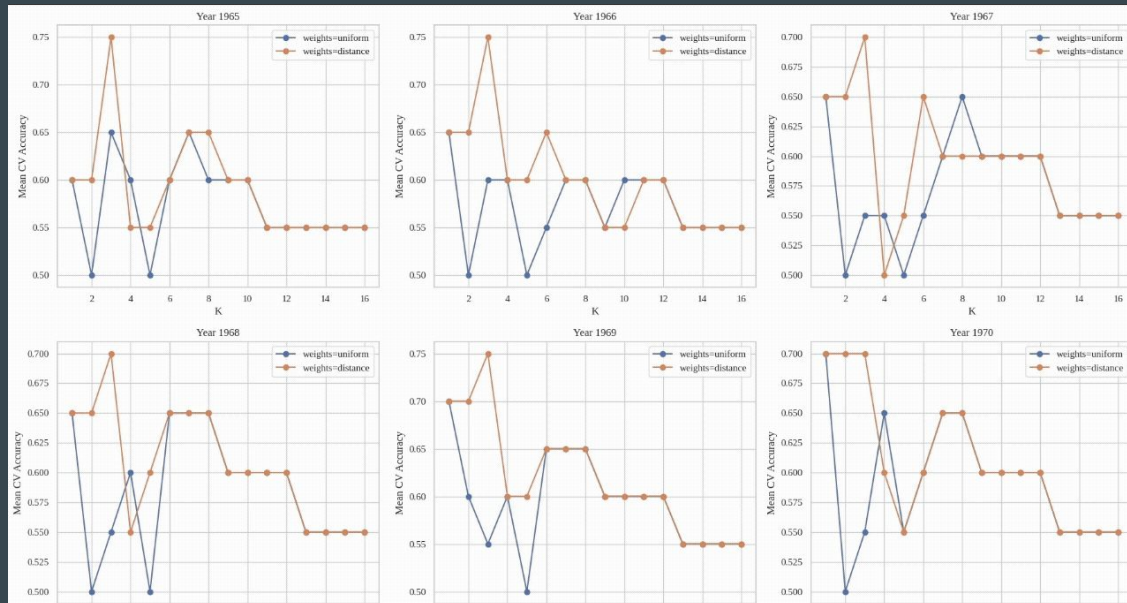
Applied KNN on the Leontief inverses computed per country per year. Model parameters found via GridSearch

Summary: Per year samples were not large enough to reliably classify regions and their economies. The model defaulted to Training on all available data led to extremely high accuracy leading to the question whether forecasting was actually possible (ie testing on unseen data). Also, explored methods of dimensionality reduction to try to combat potential overfit and maintaining robustness of euclidean distances.

Fitting A KNN Model Per Year

Visualizing the model selection, we see the average CV accuracy was never higher than ~85%. A quick visual observation shows many of these models optimal accuracy hover around ~70-75%

An optimal KNN model trained on one year of data (ex: 1977 & 2000) produces low accuracy. The sample of 25 countries is too small and the model is unable to pick up on key features to differentiate the regions. Thus, the model ends up predicting the dominant class (due to class imbalance).

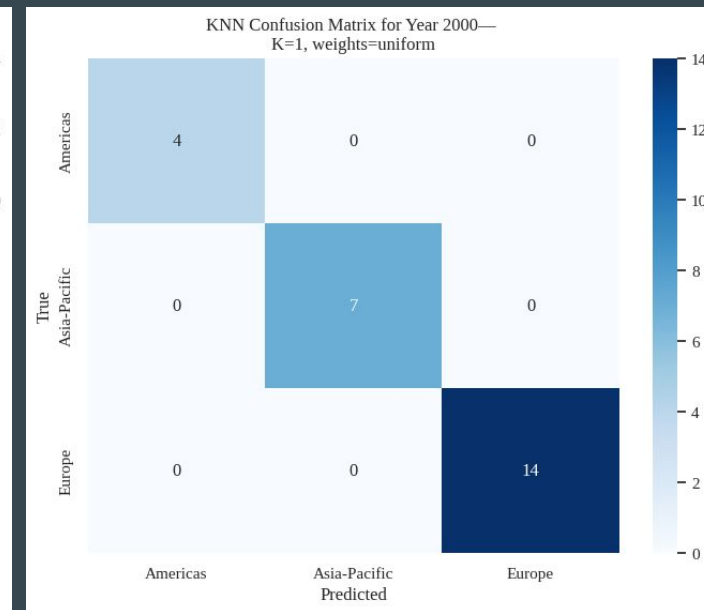
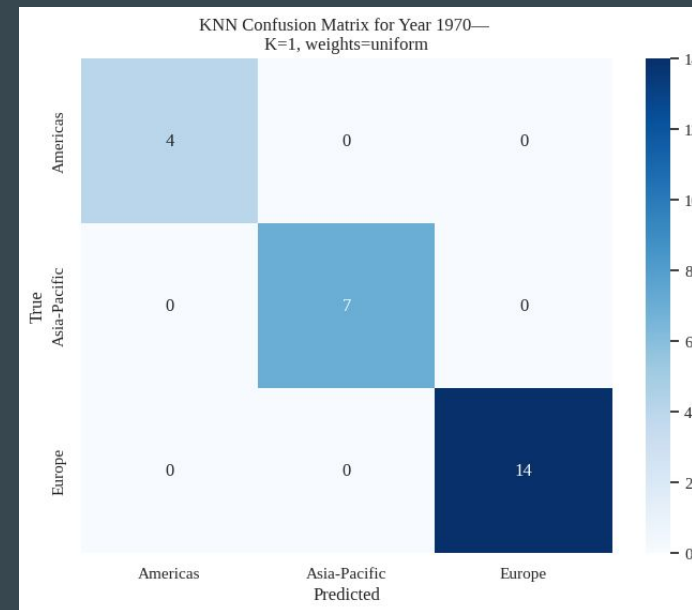
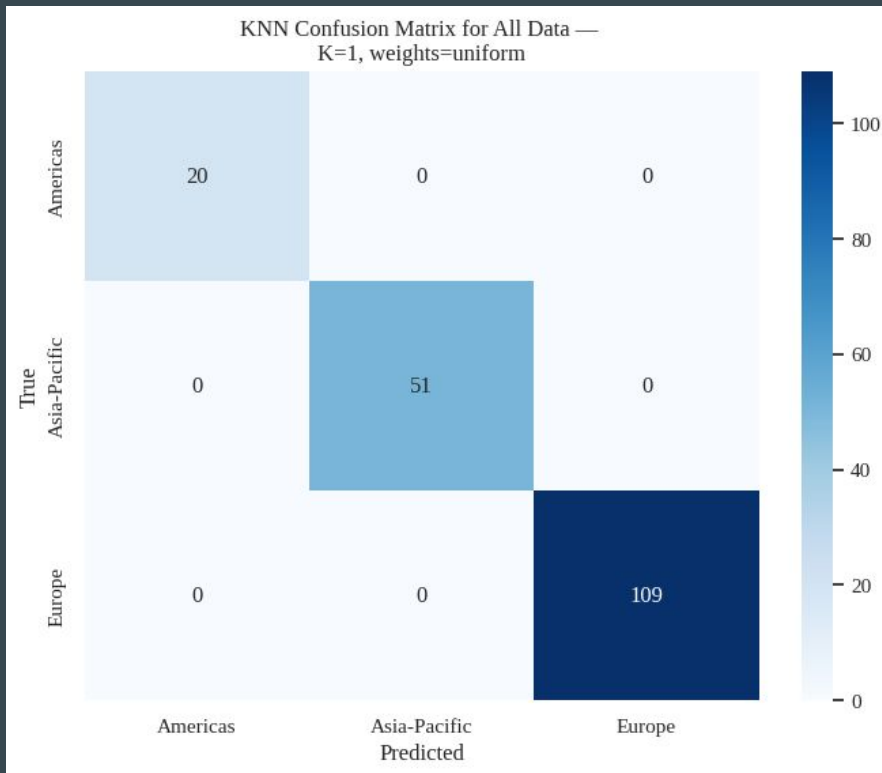


*for reference, all models trained with 80/20 train/test split

Fitting KNN on All Data

Taking all the data from (1965-2000) and training a model (80/20 train/test split), testing accuracy is perfect. Implications of structural differences per region. Questions do arise in regards to predictive power and dimensionality.

Testing on year specific data also produces perfect accuracy. (Below ex: of 1970 and 2000). More evidence to support that there are structural differences in regional economies. However we also observe low k which leads implies dimensionality issues.



Dimensionality concerns

There are 23 Features in our inverse Leontief matrix and all matrices are square to determine sector-sector interaction.

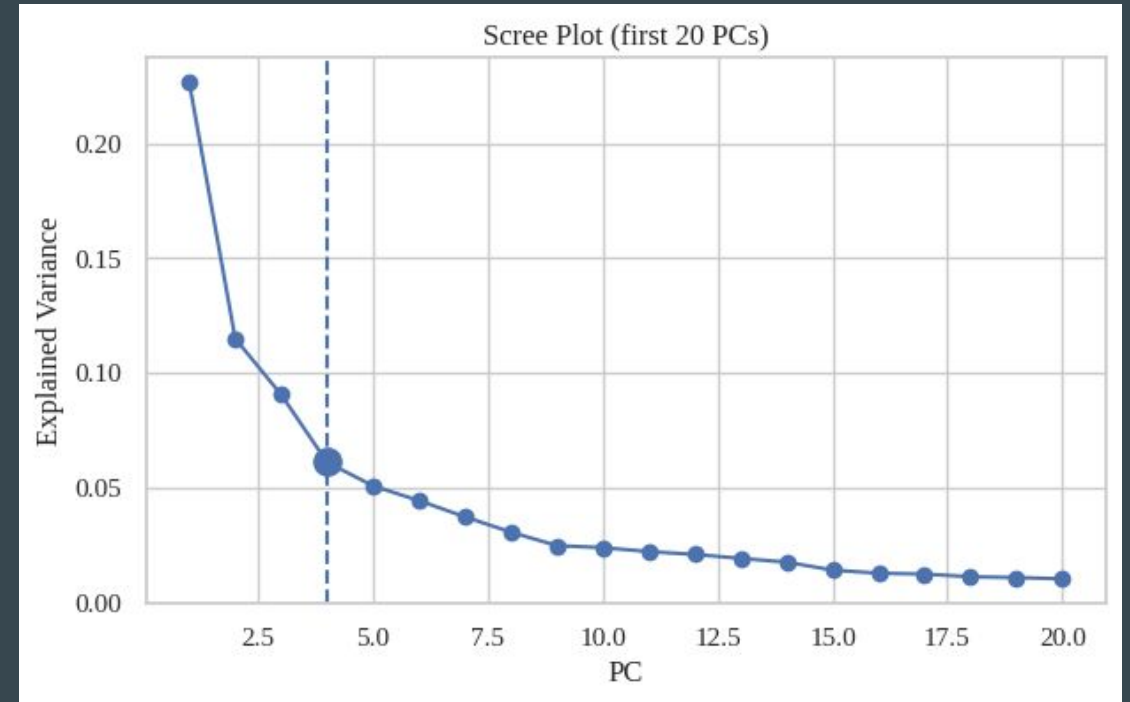
- When flattened, dimensions are 1×529
 - One country X 529 entries (23^2)
 - 25 countries implies 529 entries per year per country
- 35 years of matrices
- In such high dimensions, Euclidean distance becomes less robust as all points are “close together”.

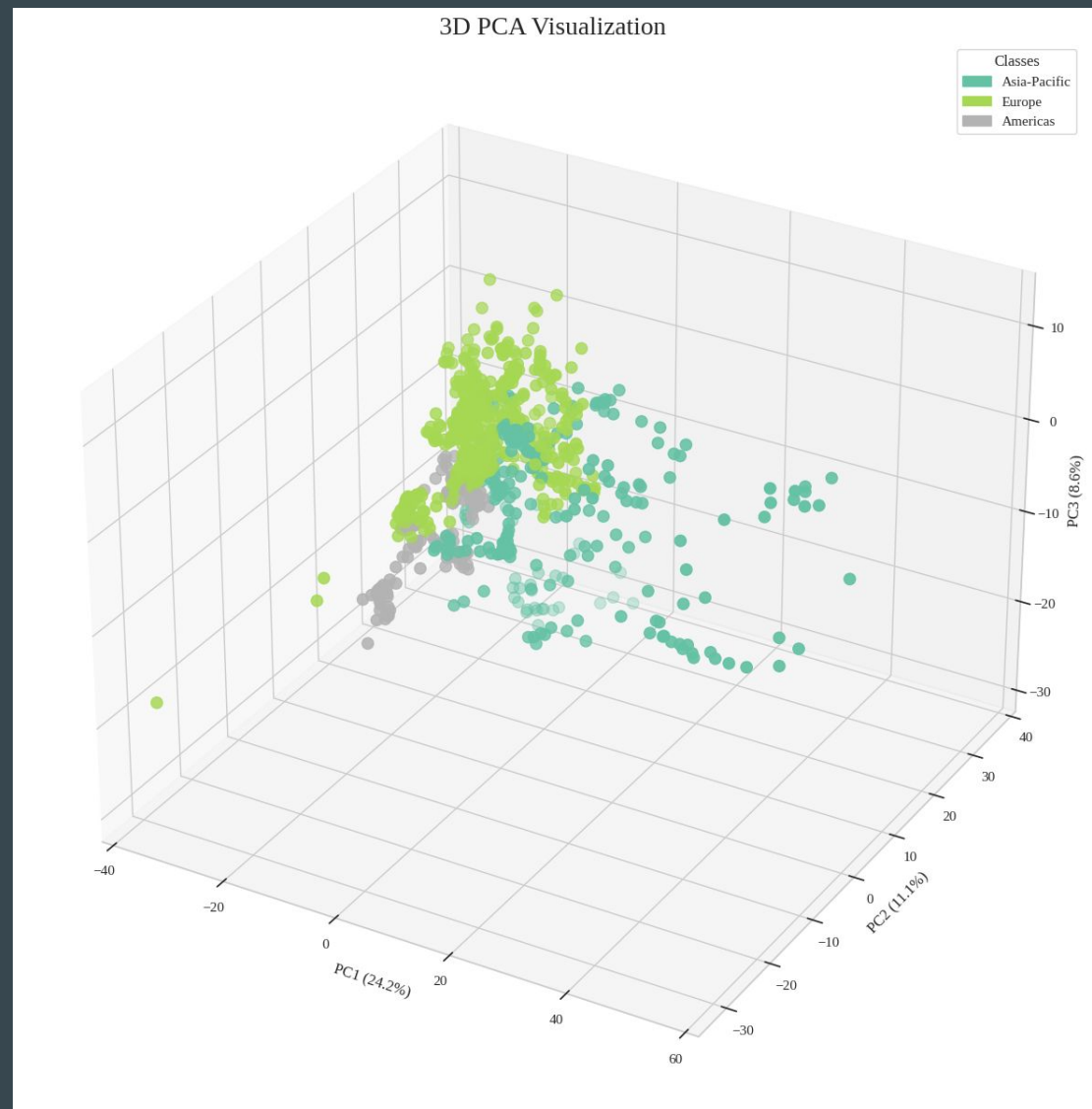
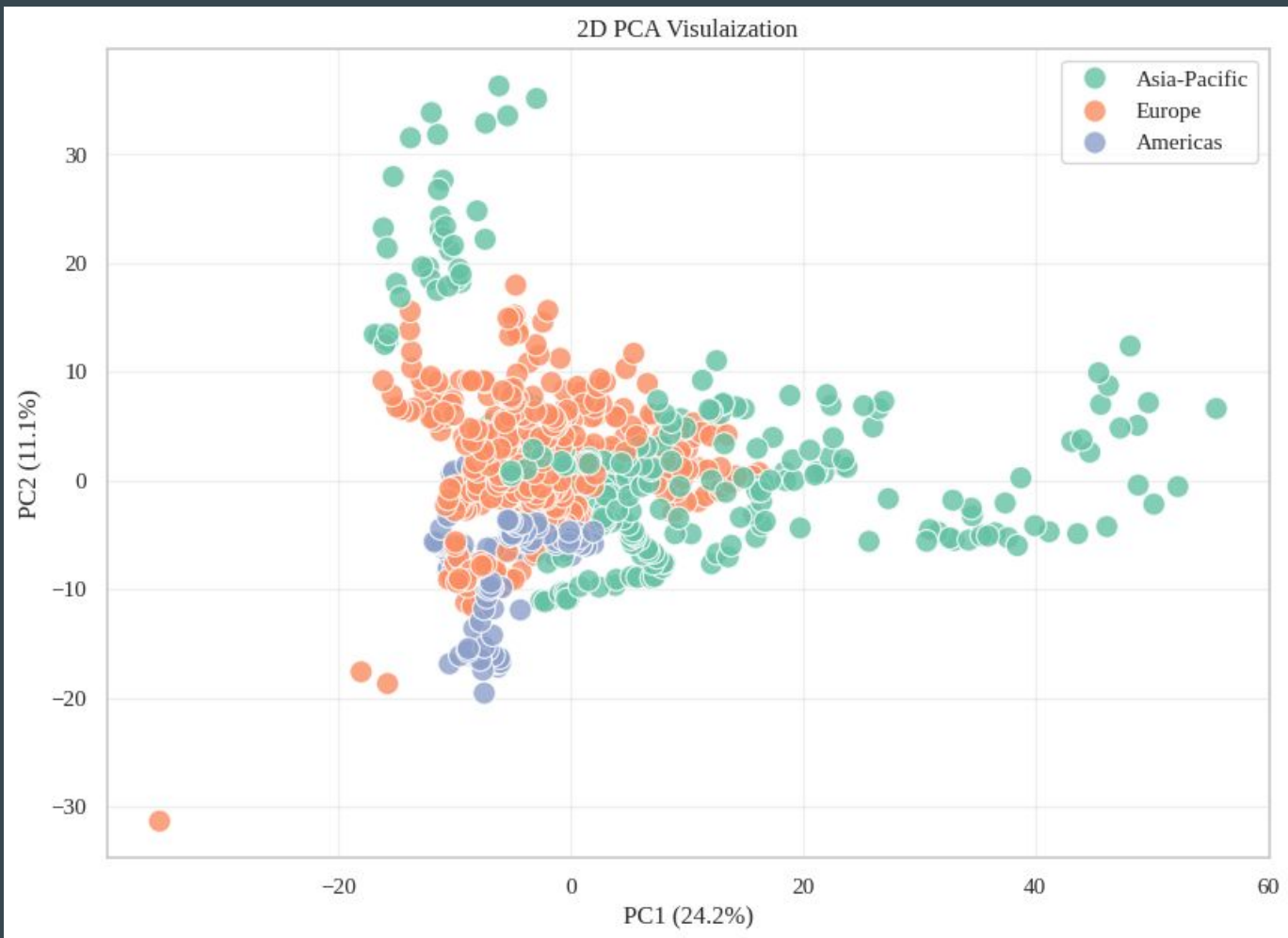
To address this, we use PCA to see if we can maintain robustness and reduce dimensionality.

PCA: All available data

Using the same Training set that was used to fit the earlier KNN model

- Plotted PCA Scree plot of explained variance captured with first 20 PC
- Using Kneedle point to help determine “elbow point” (ie: optimal PC)- 4 PC's
 - Kneedle finds the point where the function goes from steep to flat. In this case, it's capturing the most explained variance until the change in captured variance diminishes.

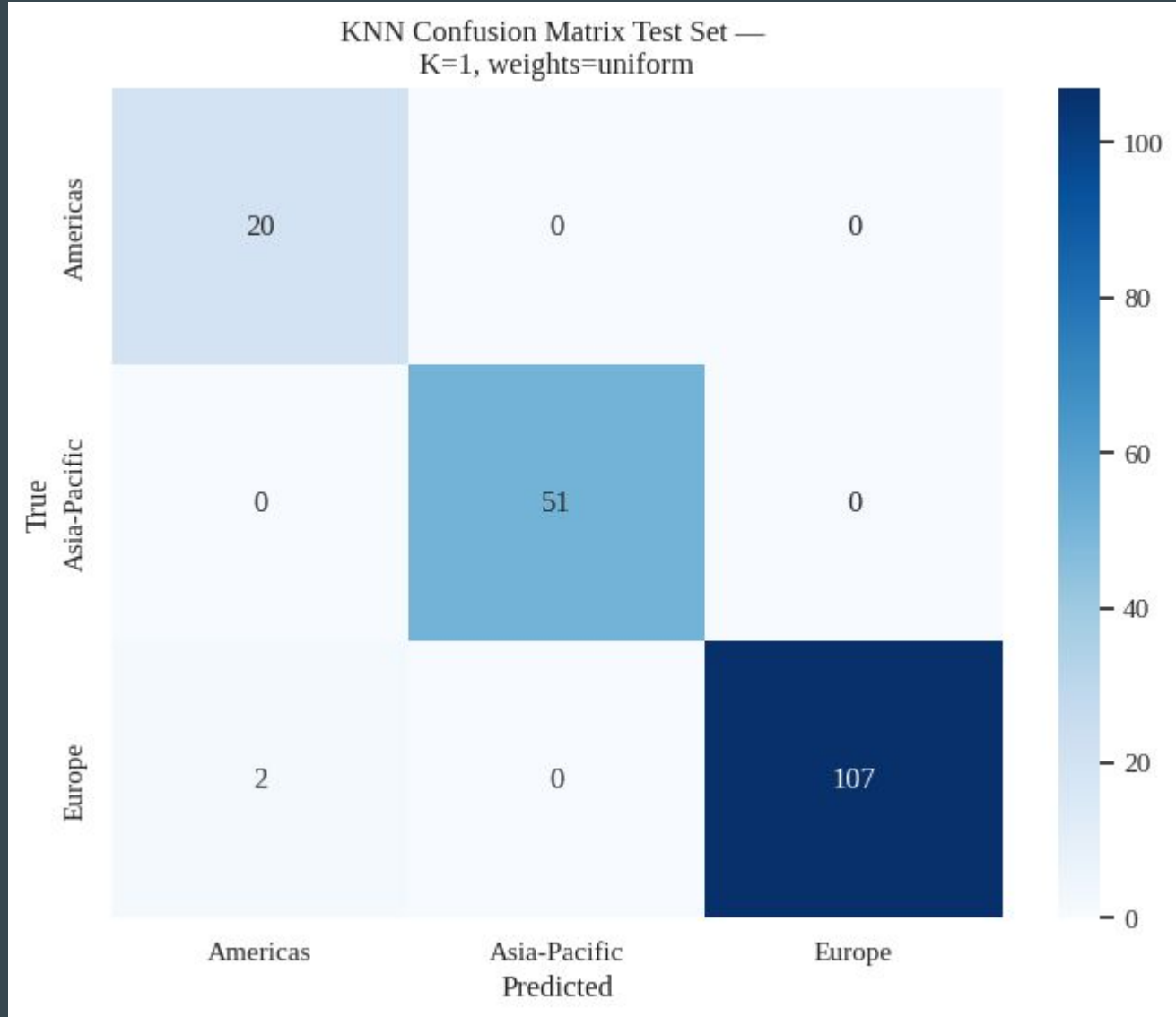




PCA Clustering Visualizations

KNN on Reduced Data

On the same test set, we see two misclassified instances. Recall, the model trained on the full dataset had no misclassification. This models' performance may have lower a misclassification but, it's only marginally worse. the consistent high performance leads me to believe that there are inherent economic differences between regions. **We will test the forecastability of the Leontief inverses.**



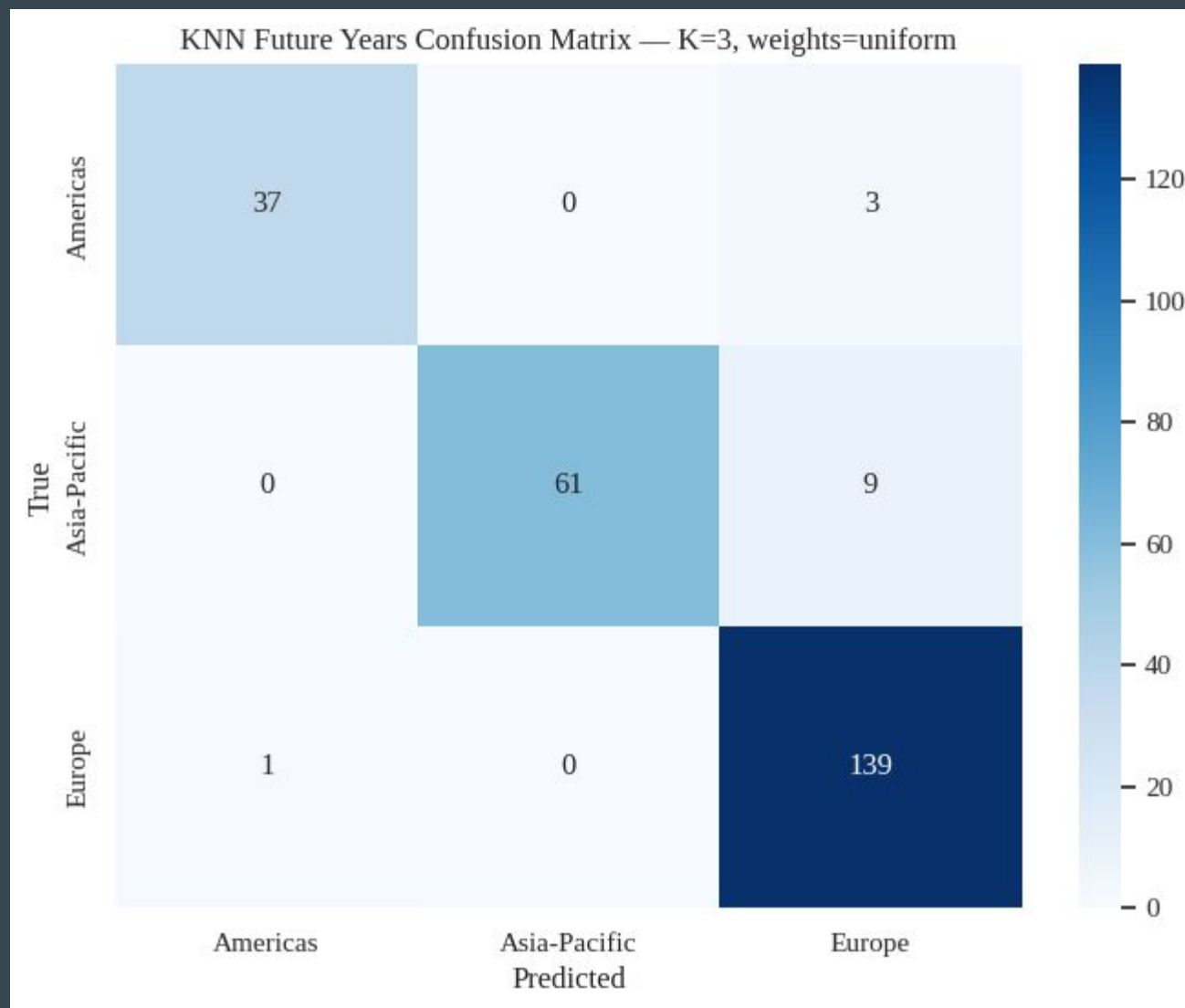
Forecastability of the Leontief Matrices

Training a KNN model on a subset of data ranging from years 1965-1990. The model will be blind to these future years and if it's still able to differentiate these regional economies it further supports the existence of structural differences.

Model fitting metrics

- Train Accuracy: 0.9985
- Test Accuracy: 0.948

A question that comes up: Is class imbalance affecting the model too heavily or are there specific instances causing misclassification?



Investigating Misclassification

Korea 1998- Foreign investment collectively pulled out of Korea leading “to a sharp contraction of economic... a negative 6.7 percent growth, the worst in modern Korean history.”.

Taiwan 1990’s- Massive political overturning and democratization could have be a potential explanation. (Key sectors to look for may be government spending metrics).

Canada Late 1990’s- Canada was freshly out of a recession (early 1990’s) and typically, these economies post recession experience a peak or “boom” (expansionary stage) before stabilizing to a more consistent level state. (lifecycle of economies)

Greece 1999- Greece began implementing labor laws regarding foreign nationals. Also, their nation’s debt is relatively high which could affect the sector to sector behavior of the economy.

Misclassified future-year cases:

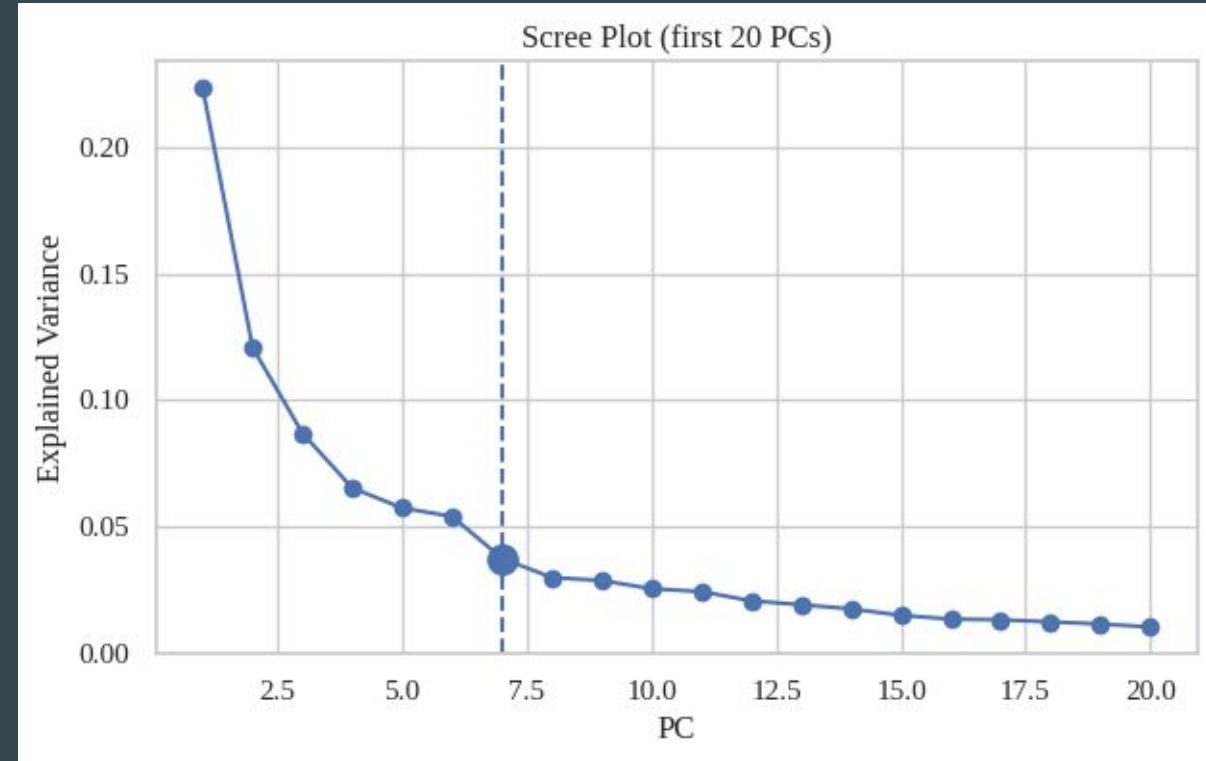
	Country	Year	True Region	Predicted Region
0	CAN	1995	Americas	Europe
1	CAN	1996	Americas	Europe
2	CAN	2000	Americas	Europe
3	GRC	1999	Europe	Americas
4	KOR	1998	Asia-Pacific	Europe
5	TWN	1993	Asia-Pacific	Europe
6	TWN	1994	Asia-Pacific	Europe
7	TWN	1995	Asia-Pacific	Europe
8	TWN	1996	Asia-Pacific	Europe
9	TWN	1997	Asia-Pacific	Europe
10	TWN	1998	Asia-Pacific	Europe
11	TWN	1999	Asia-Pacific	Europe
12	TWN	2000	Asia-Pacific	Europe

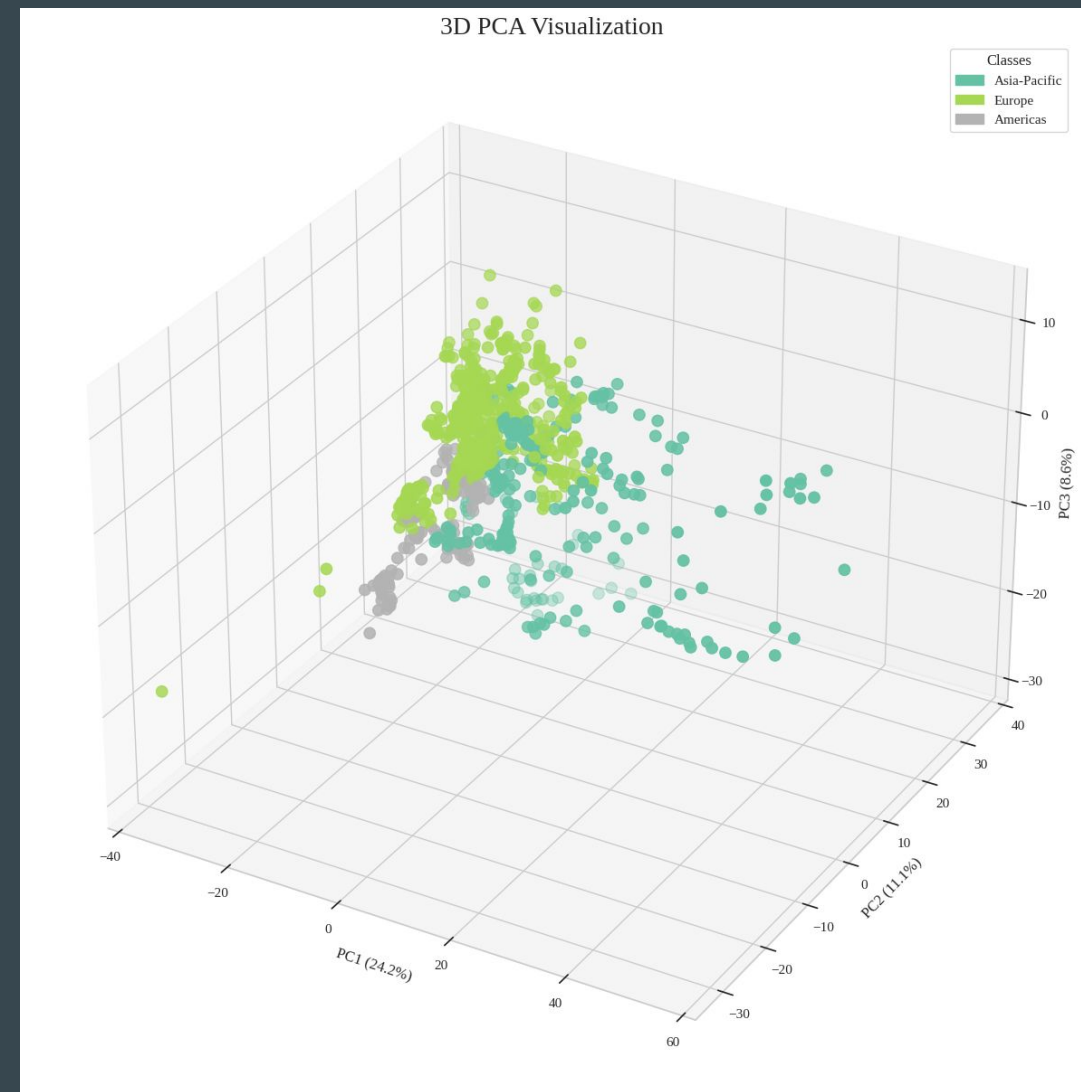
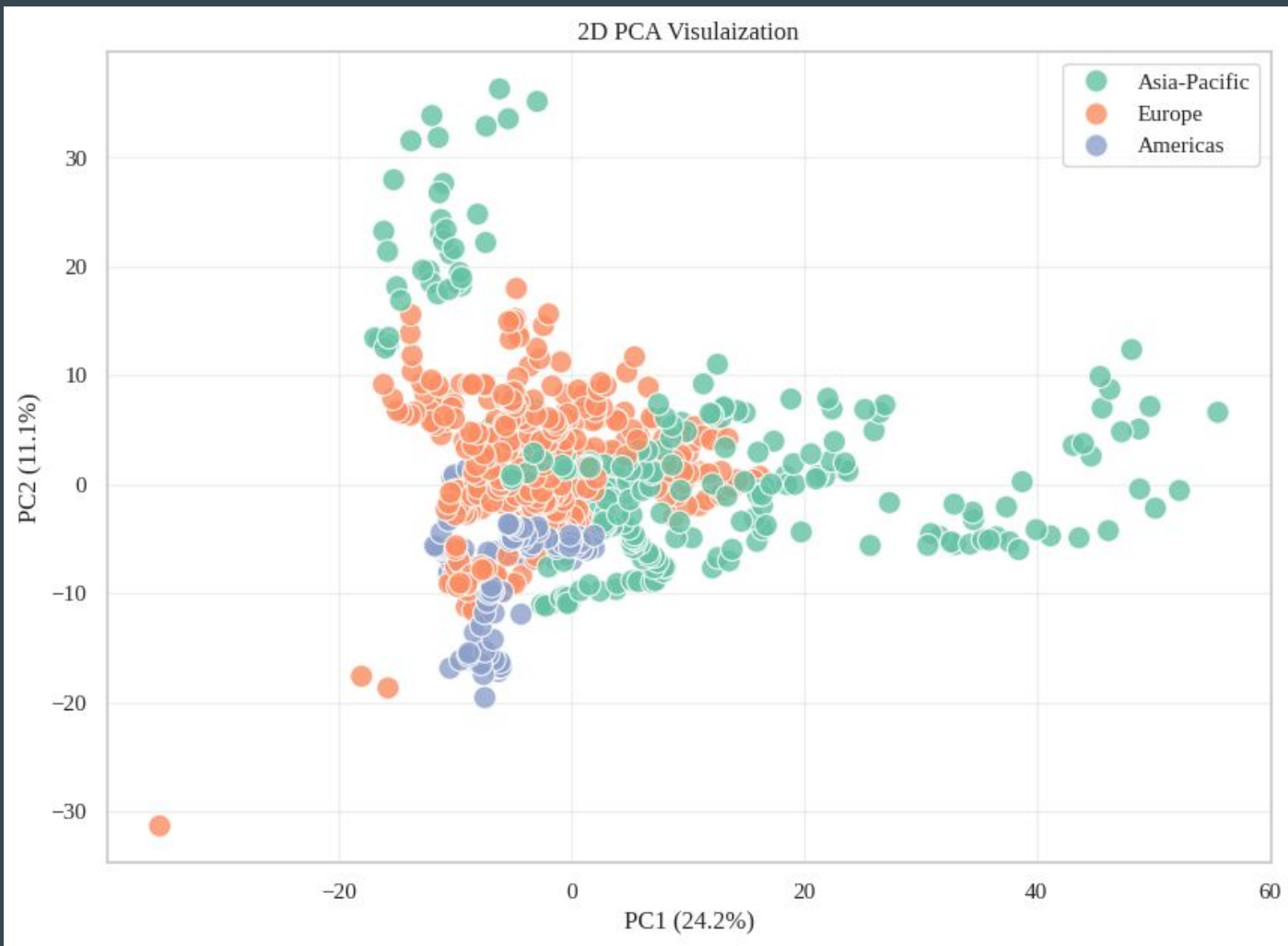
PCA: Subsetted Years

To address the same potential dimensionality concerns,

Using the same Training set that was used to fit the earlier KNN model

- Plotted PCA Scree plot of explained variance captured with first 20 PC
- Again, used Kneedle point to help determine “elbow point” (ie: optimal PC)- 7 PC's





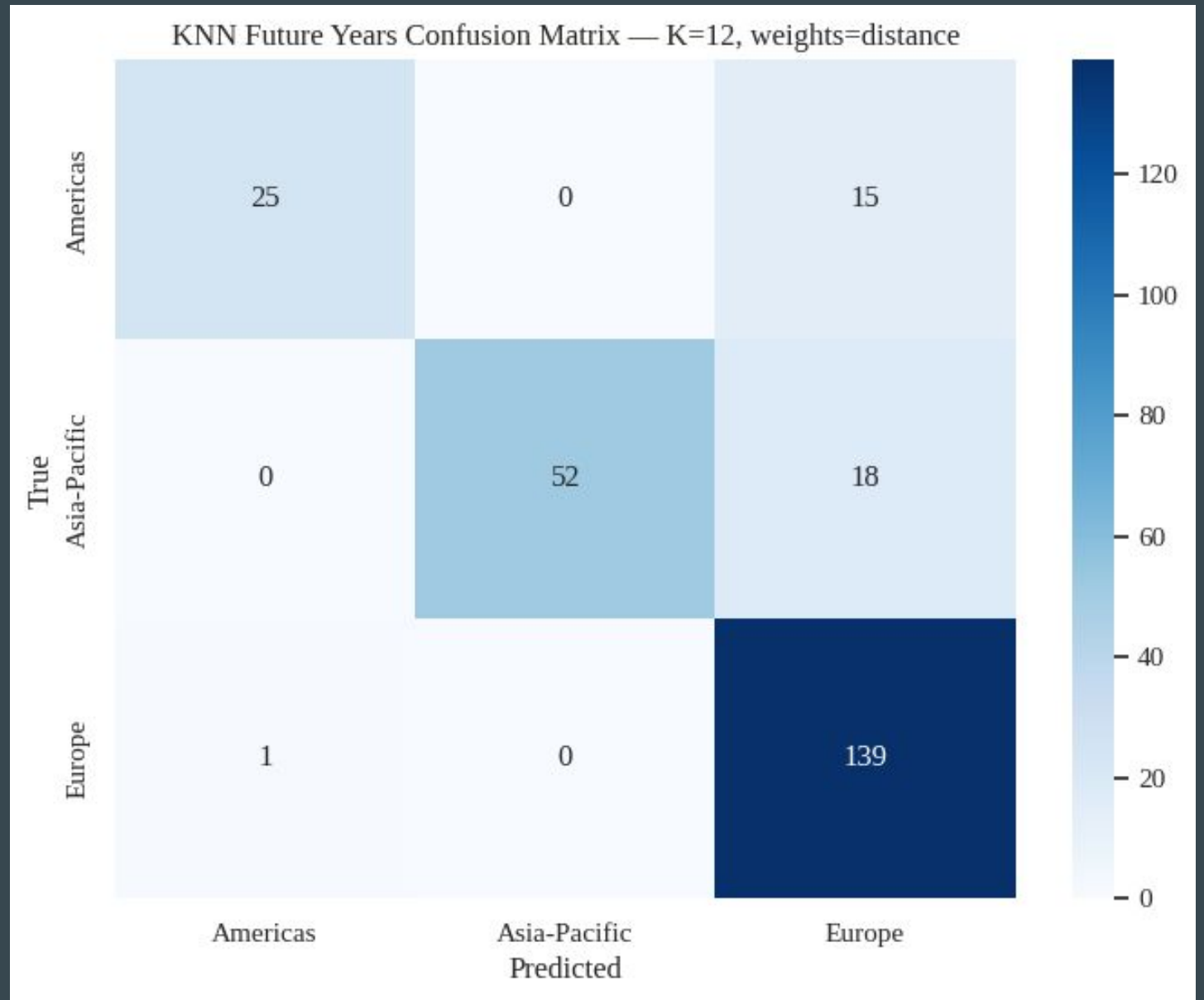
PCA Clustering Visualizations

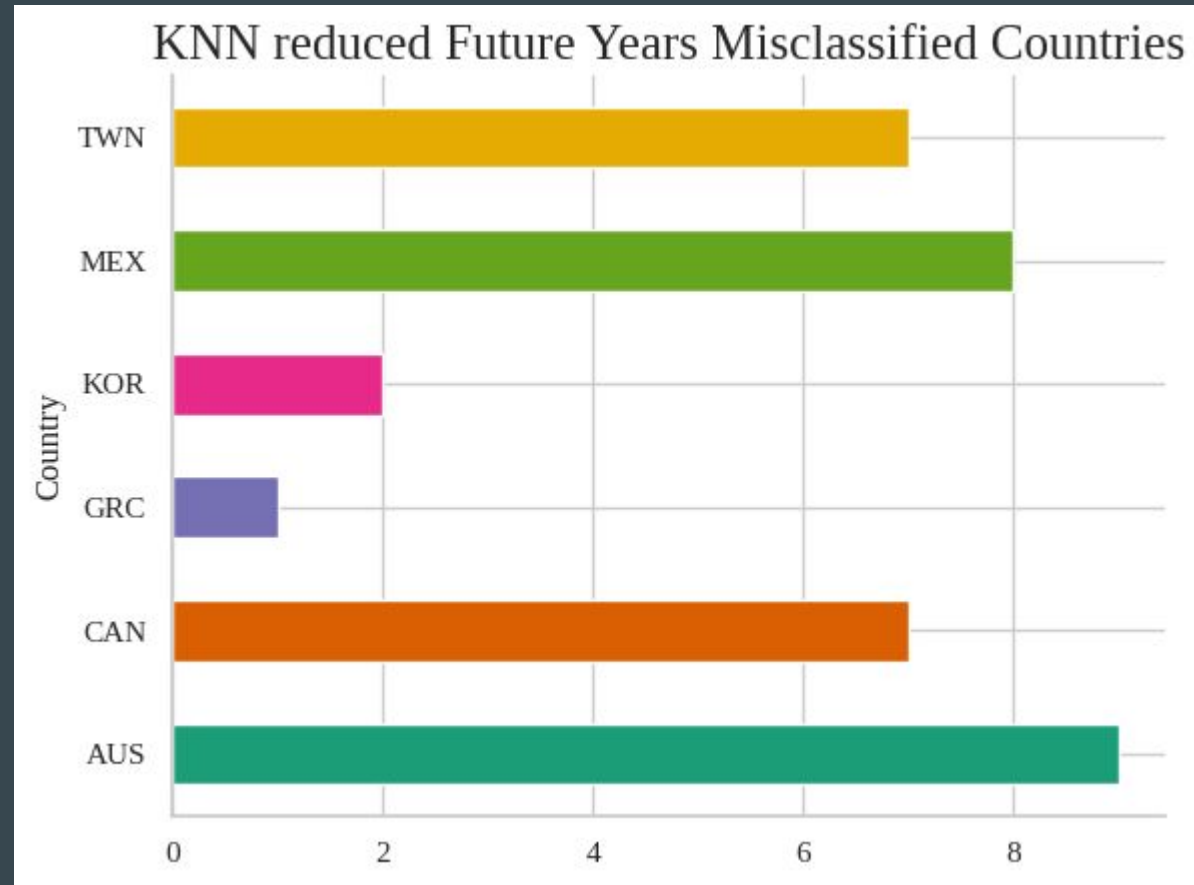
Reduced KNN on Future Years

Metrics for the model

- Train Accuracy: 1.0
- Test Accuracy: 0.864

Poorly performing after dimensionality reduction. The high train and low test accuracy tells me PCA causes key variance to be lost in the model, thus losing it's predictive power. You can also see the overpowering prediction of the Europe class.





Lost too much important information and is unable to classify countries well, ended up heavily classifying based on class imbalance.

index	Country	Year	True Region	Predicted Region
0	AUS	1992	Asia-Pacific	Europe
1	AUS	1993	Asia-Pacific	Europe
2	AUS	1994	Asia-Pacific	Europe
3	AUS	1995	Asia-Pacific	Europe
4	AUS	1996	Asia-Pacific	Europe
5	AUS	1997	Asia-Pacific	Europe
6	AUS	1998	Asia-Pacific	Europe
7	AUS	1999	Asia-Pacific	Europe
8	AUS	2000	Asia-Pacific	Europe
9	CAN	1994	Americas	Europe
10	CAN	1995	Americas	Europe
11	CAN	1996	Americas	Europe
12	CAN	1997	Americas	Europe
13	CAN	1998	Americas	Europe
14	CAN	1999	Americas	Europe
15	CAN	2000	Americas	Europe
16	GRC	1999	Europe	Americas
17	KOR	1994	Asia-Pacific	Europe
18	KOR	1995	Asia-Pacific	Europe
19	MEX	1993	Americas	Europe
20	MEX	1994	Americas	Europe
21	MEX	1995	Americas	Europe
22	MEX	1996	Americas	Europe
23	MEX	1997	Americas	Europe
24	MEX	1998	Americas	Europe
25	MEX	1999	Americas	Europe
26	MEX	2000	Americas	Europe
27	TWN	1994	Asia-Pacific	Europe
28	TWN	1995	Asia-Pacific	Europe
29	TWN	1996	Asia-Pacific	Europe
30	TWN	1997	Asia-Pacific	Europe
31	TWN	1998	Asia-Pacific	Europe
32	TWN	1999	Asia-Pacific	Europe
33	TWN	2000	Asia-Pacific	Europe

Takeaway

There are inherent regional differences in economic structure.

- This is seen in our fairly high success in classifying regions based on Leontief inverses

However, the differences are very likely due to very specific industry and sectoral composition.

- We see this as PCA generally reduces the performance metrics of our model and it's classification.
 - Since PCA generalizes the variance of features to reduce dimensions, we lose the exact/key signals that allow for differentiating regions.

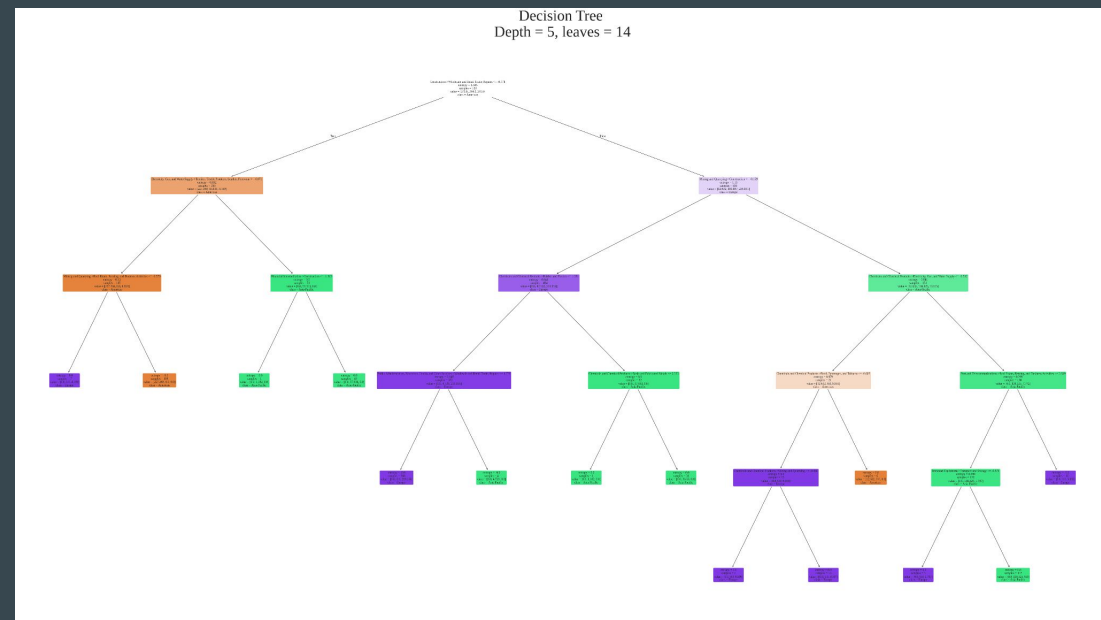
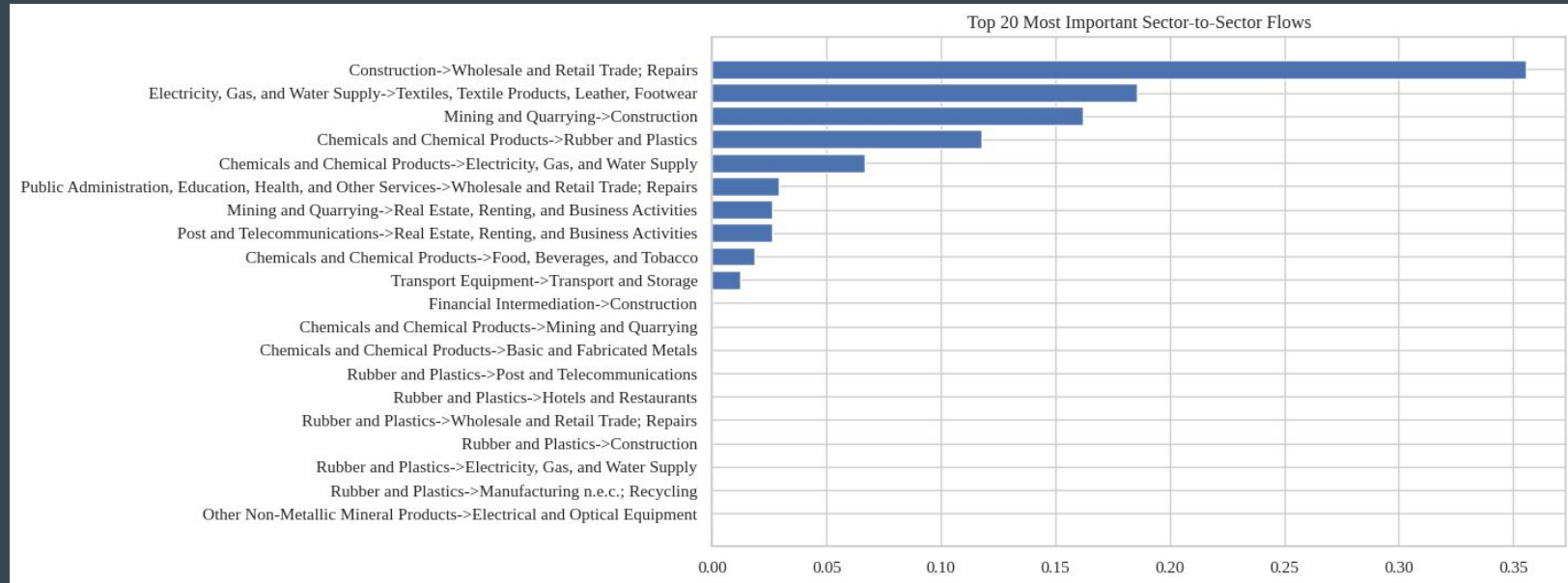
Decision Tree

For interpretability, implemented a Decision Tree on all available data to visually inspect the specific sector -> sector interactions that differentiate the regions.

The tree seems somewhat complex but when inspecting the fitting, we see the train and test accuracy are relatively high/similar.

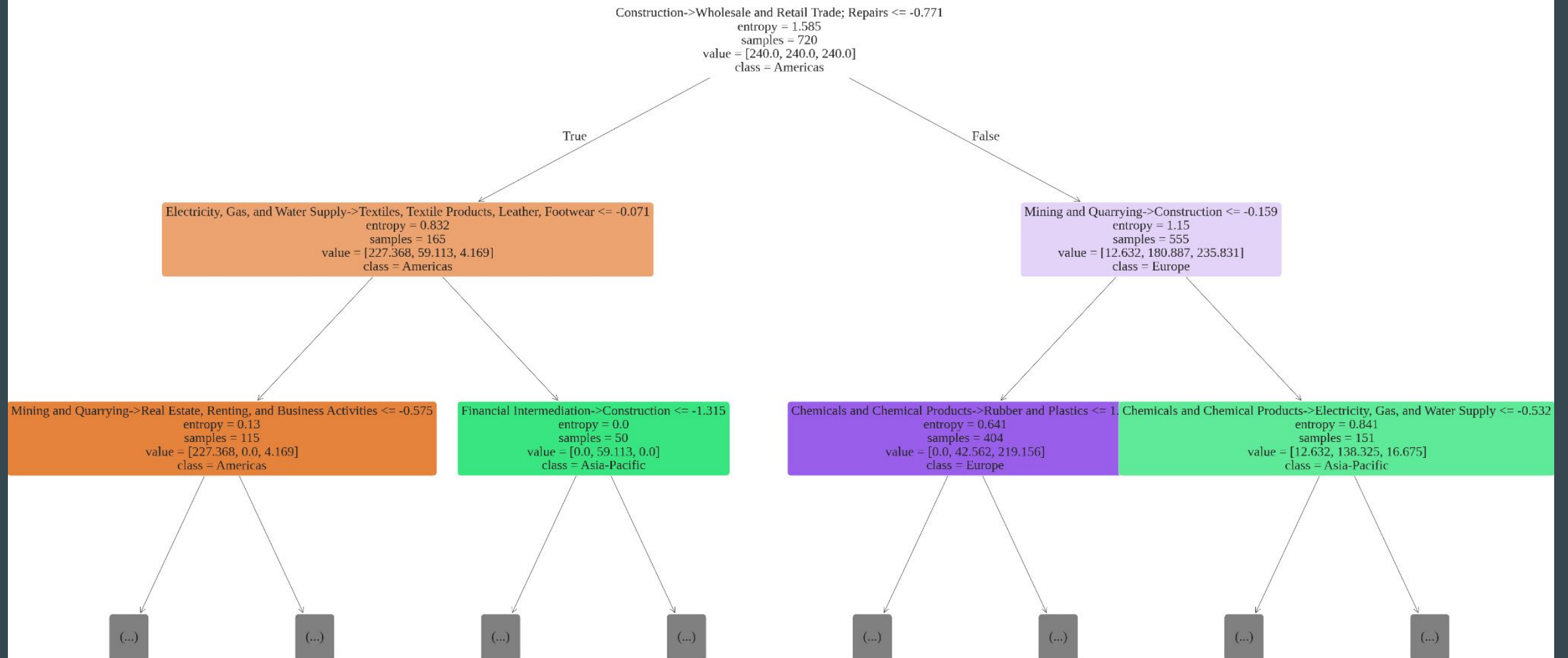
- Train Accuracy: 1.0
- Test Accuracy: 0.9833333333333333

This indicates a relatively well-fit model that captures the key structural relationships between sectors.

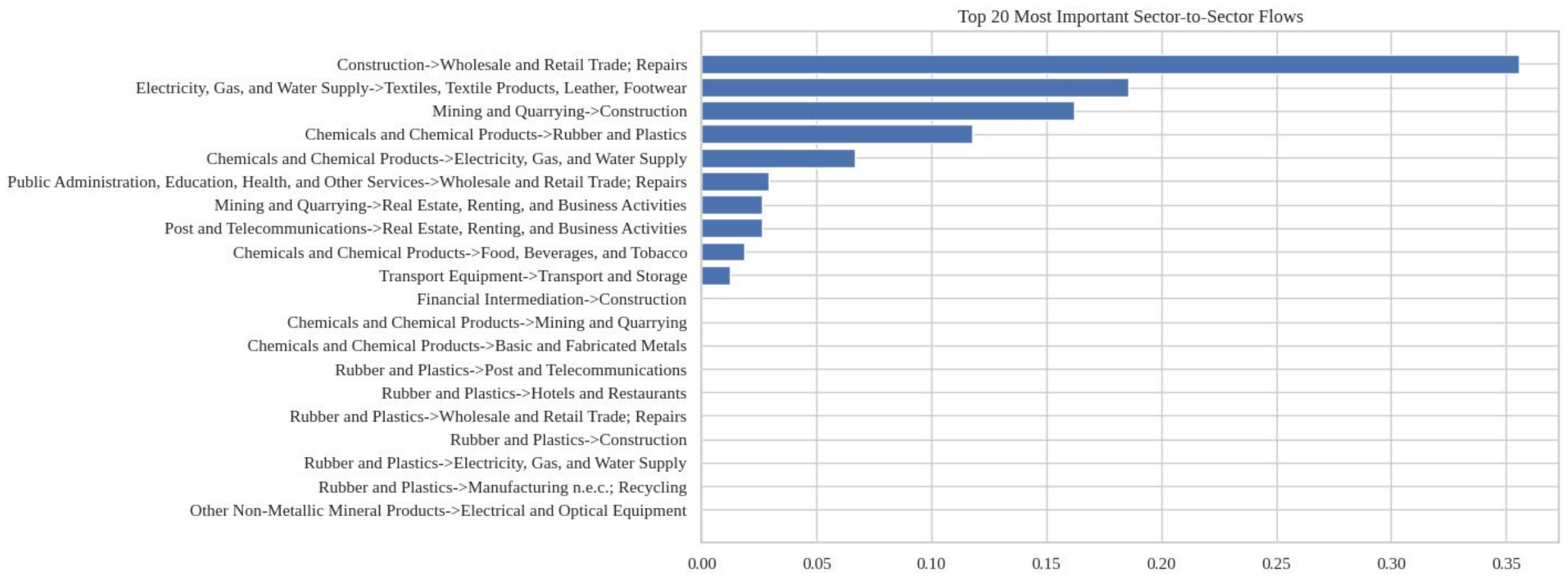


Decision Tree

Depth = 5, leaves = 14



First 2 levels of the Decision Tree



Feature Importance chart of the Decision Tree

Feature importances are computed as:

- Shows how much each feature contributes to reducing impurity (e.g., Gini).
- Calculated by summing the impurity decrease from all splits using that feature.
- Higher values implies more influence on the tree's decisions.

Try unsupervised Clustering to see if we obtain similar results

K-Means Clustering

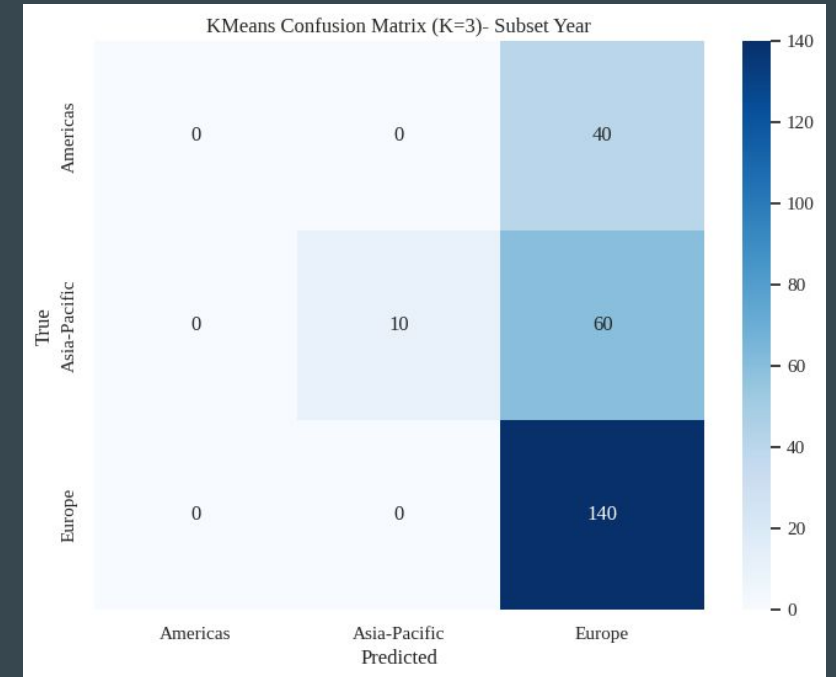
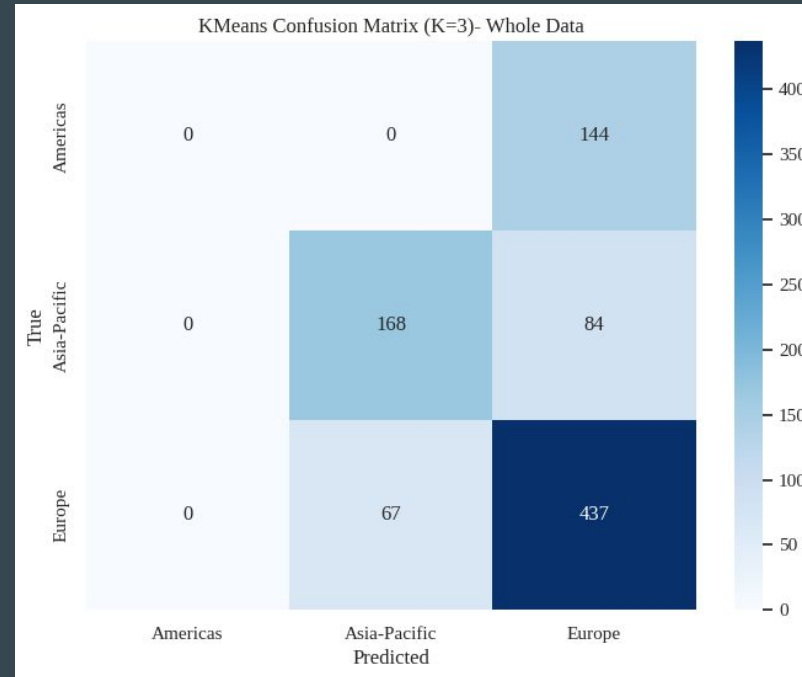
We set $K=3$ to capture meaningful representation of the three regions: Europe, Asia-Pacific, and Americas.

Implemented K-Means Clustering on:

- All available data
- The Forecasting subset

To evaluate cluster performance:

- Assigned each cluster a label based on a majority vote of countries within the cluster.
- Used confusion matrices to assess how well clusters correspond to actual regional classifications.

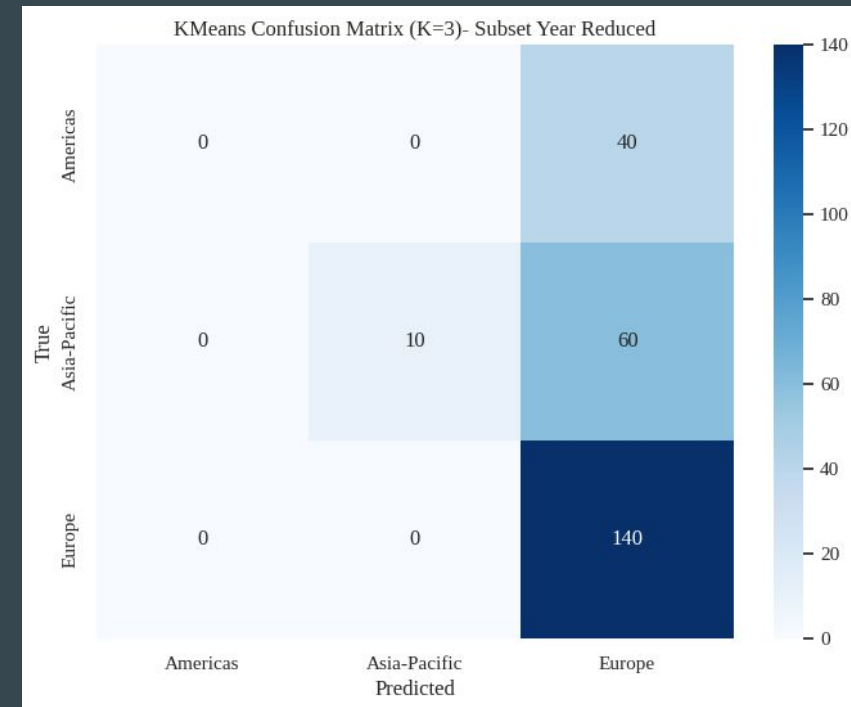


The models show strong class imbalance, heavily favoring predictions for Europe while being largely insensitive to the Americas class.

Clustering with Reduced Dimensions

Using the same optimal principal components identified in earlier models, we refit the clustering.

However, the clustering once again does not show strong separation among all three regions, unlike the clearer distinctions observed with the KNN algorithm



Analysis

Raw clustering into three groups shows limited sensitivity, which is expected given the nature of the aggregation/grouping of clusters. This doesn't make separation impossible, but it may require finer breakdowns (e.g., East Asia, Eurasia, Western Europe, North America).

- Data limitations highlighted in our EDA constrain the ability to achieve clean separation across all three regions.
- Theoretically, if the three (Europe, Asia-Pacific, Americas) regions **could** be well-separated, class imbalance effects should be minimal, as the three classes would naturally separate and cluster together.
- Nonetheless, there's some visible separation between Asia-Pacific and Europe.

A potential direction is to explore methods that differentiate East (Asia-Pacific) from West (Americas & Europe), possibly by using targeted feature engineering.

Some exploration

To further investigate, we use silhouette scores to determine an optimal number of clusters (K) for our data.

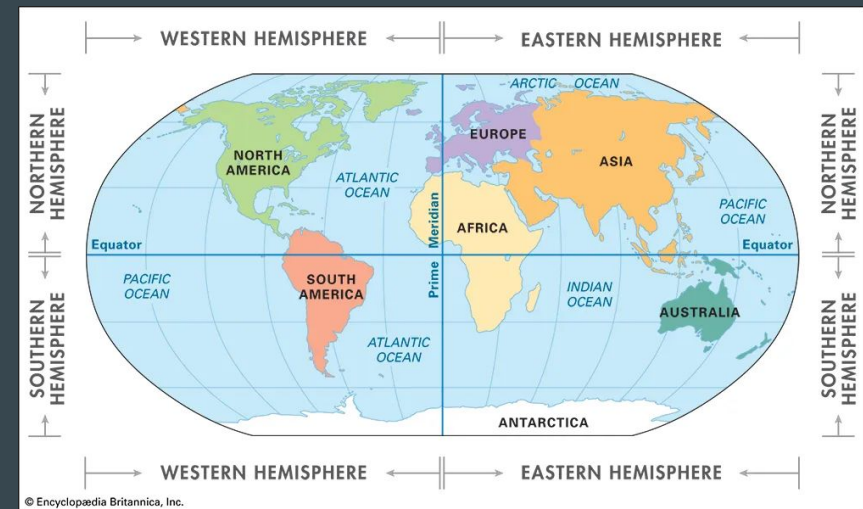
- Optimal K by silhouette: 2
- This supports the idea that the data may naturally split into East vs. West regions rather than three distinct groups.

Geopolitical considerations:

- Australia is geographically in the East but often classified as “Western” in political science; we account for both perspectives in the clustering (ie fitting two different models, geographically and politically).
- South America, despite being a separate continent, historically has high European influence and settlement; it will be treated as part of the Western group.

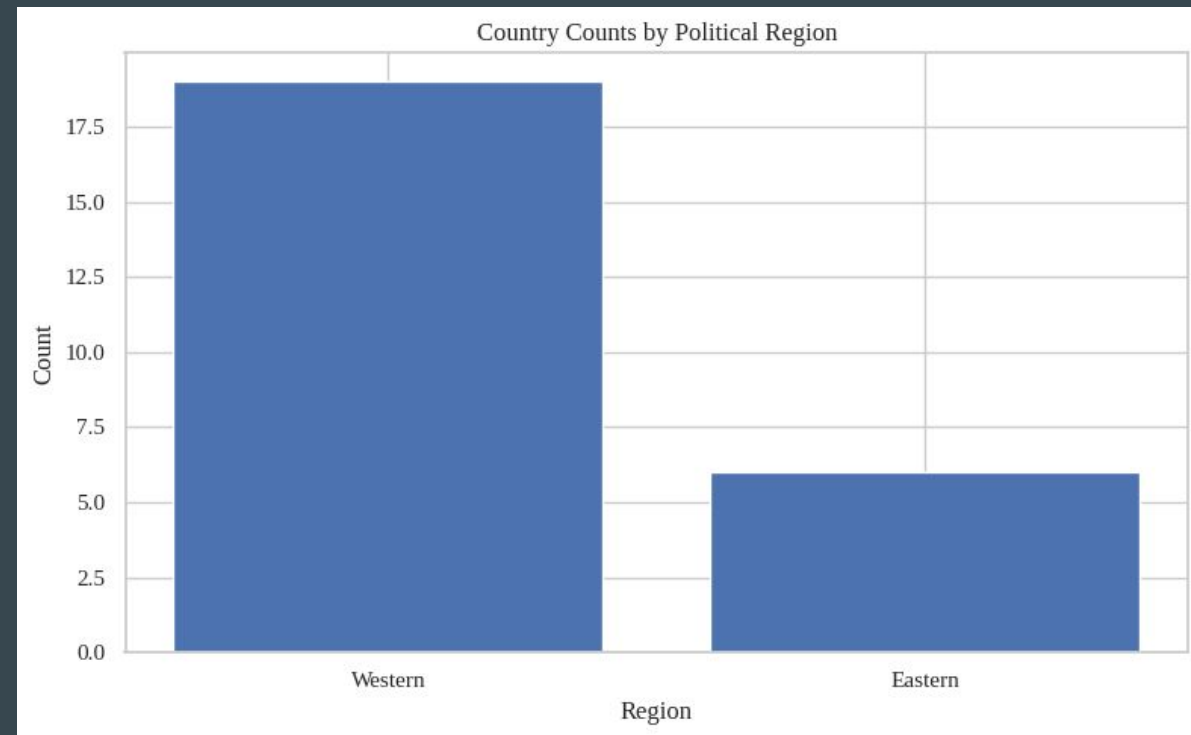
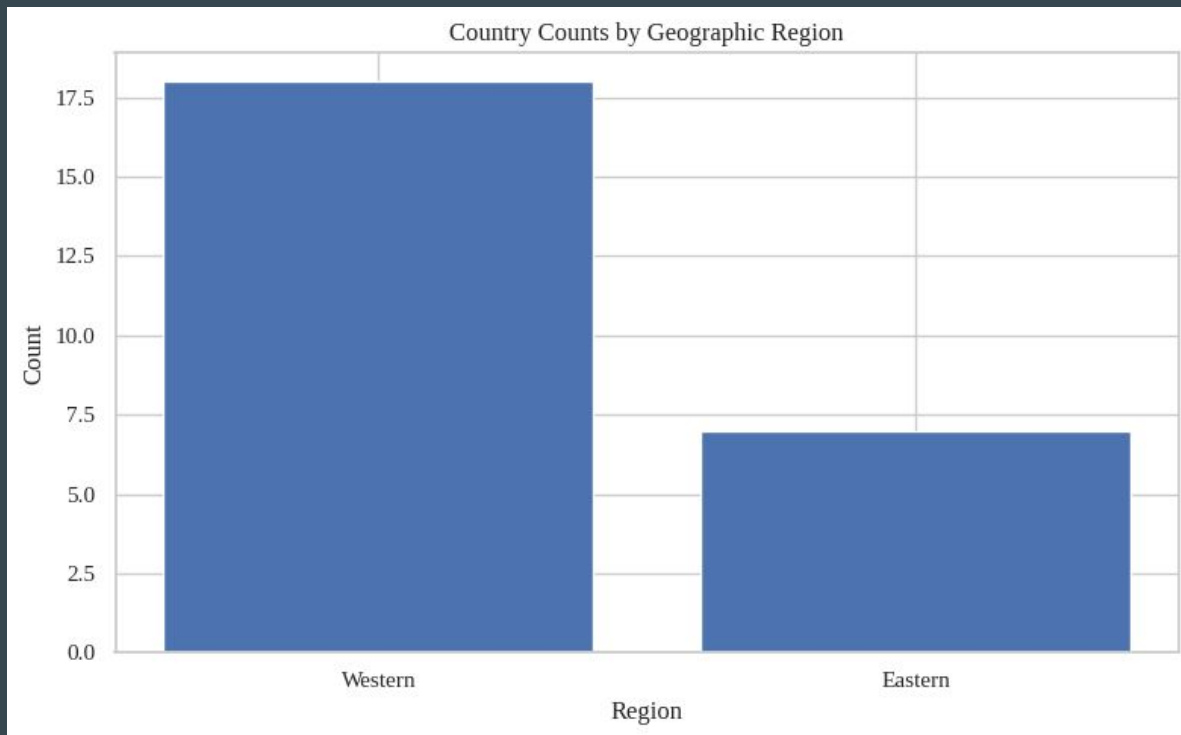
The silhouette score measures how similar a point is to its own cluster compared to the nearest other cluster. Values close to 1 indicate well-separated clusters, while values near 0 suggest overlapping clusters.

To optimize this metric, we focus on broader structural patterns and set aside the finer distinctions captured by KNN, **where the regional differences explored earlier are likely very specific and nuanced.**



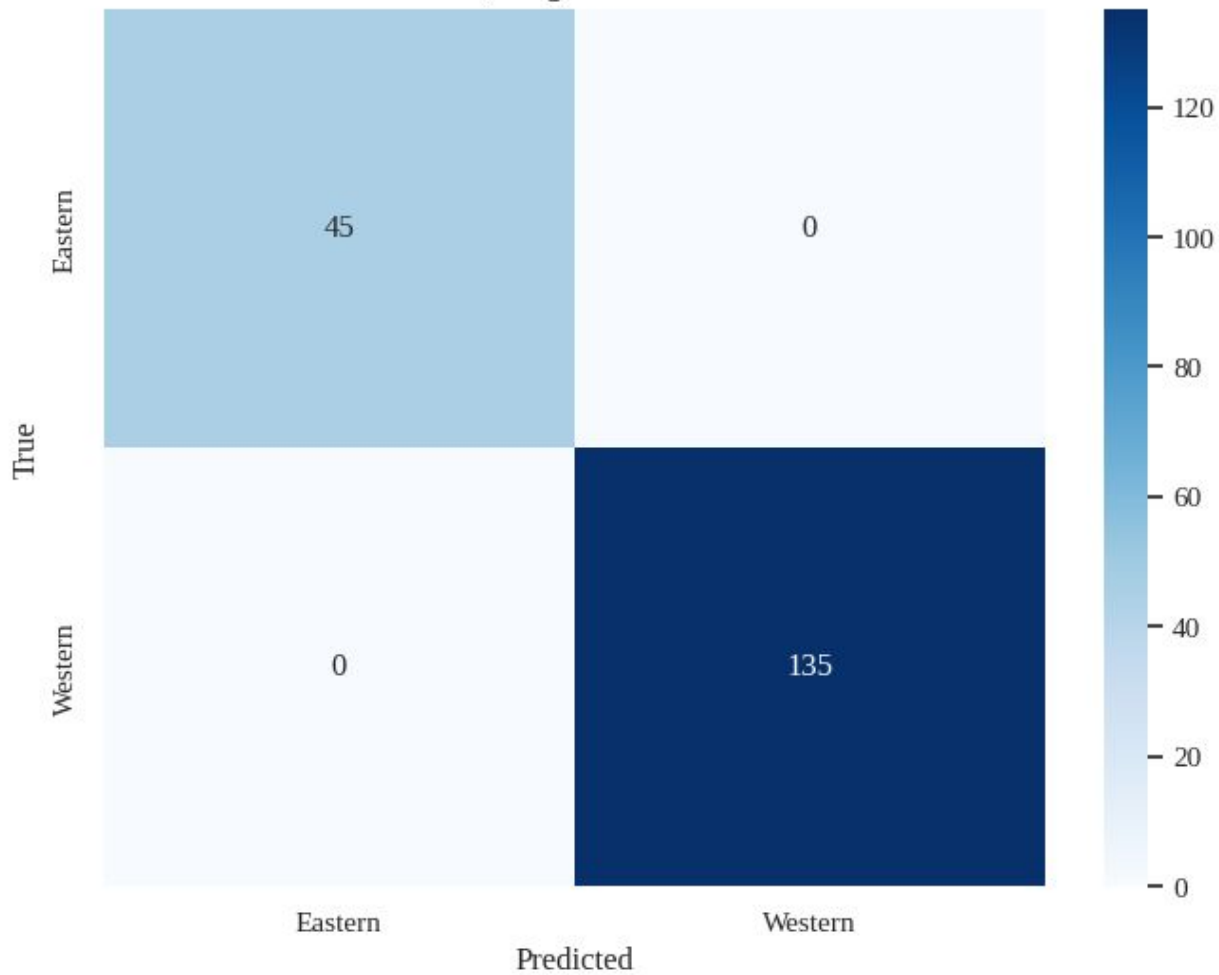
We used this newly structured data and essentially redid the earlier classification (with East/West labels). The following only contains some of the new and interesting results.

Generally, KNN did well, forecasting seems to be affected the most by class imbalance.
More details are in our Notebook

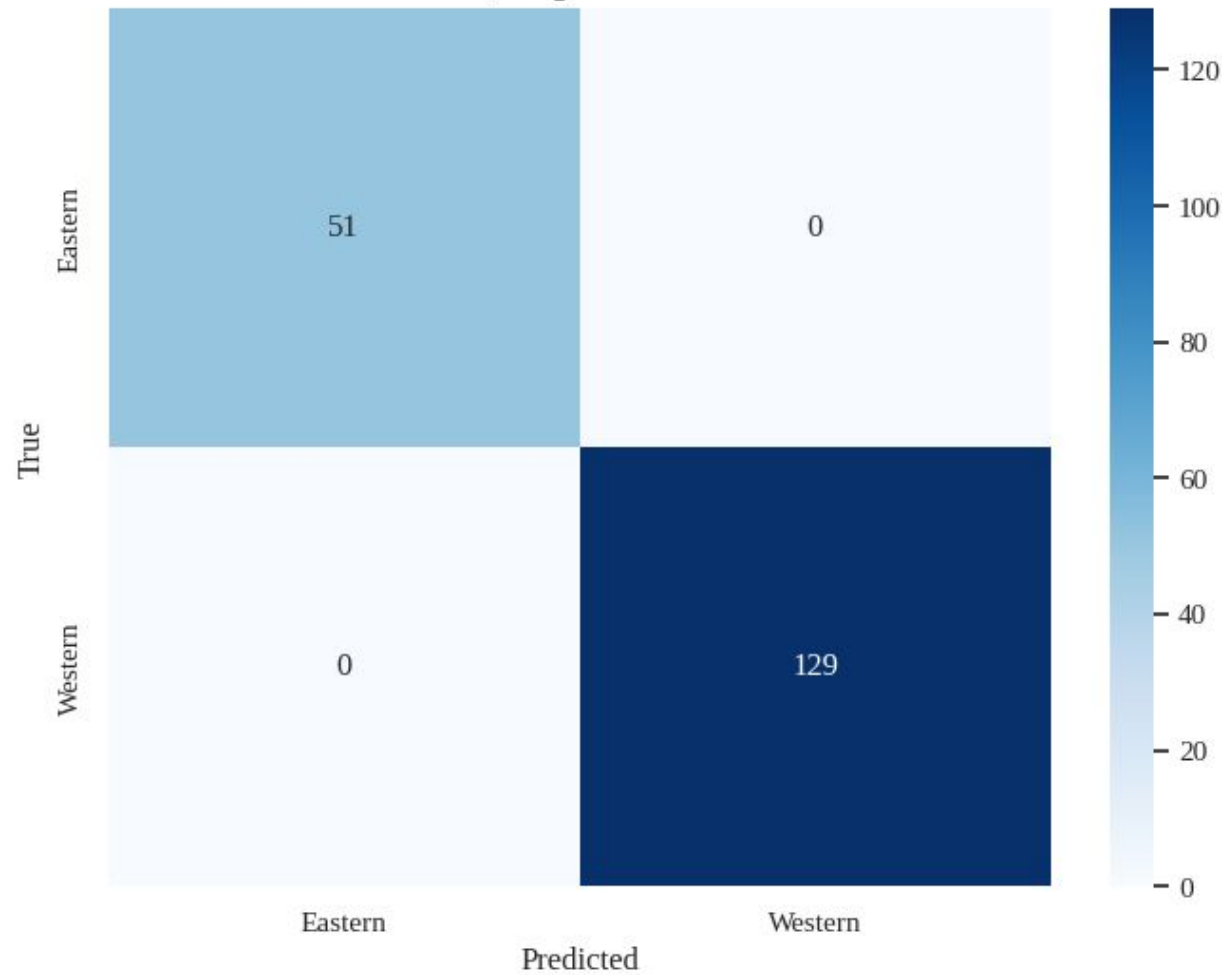


Classes EDA- Definitely class imbalance which may impact our model.

KNN Confusion Matrix for All Data — political
K=1, weights=uniform



KNN Confusion Matrix for All Data — geographic
K=1, weights=uniform

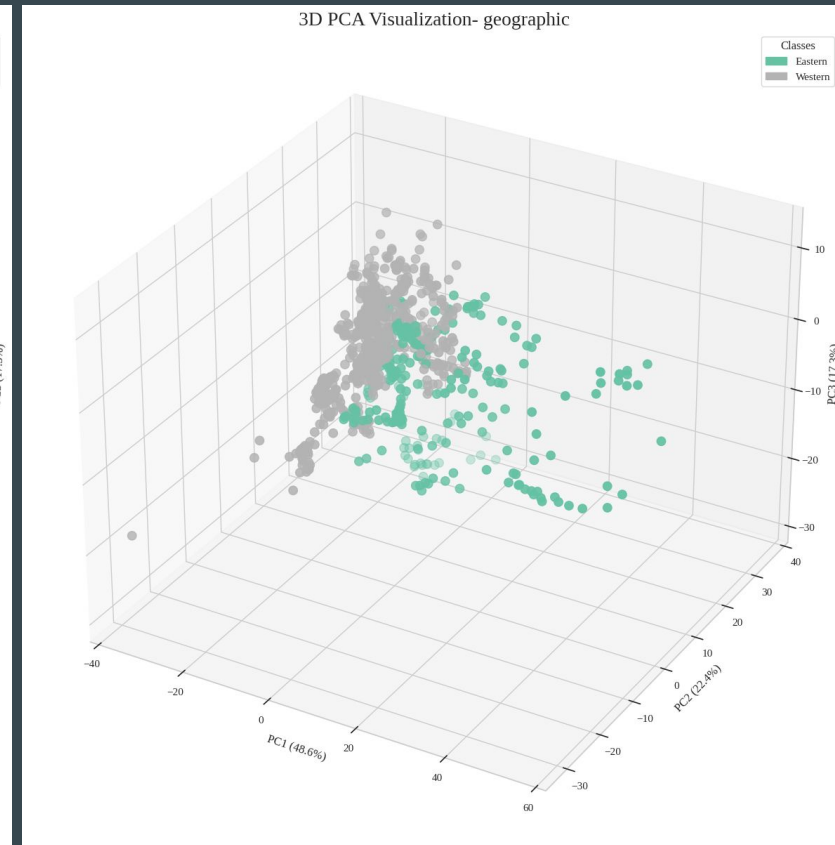


EDA on Robustness of New Classes - Very sensitive

New Result

PCA is highly effective at separating these new classes, unlike the finer distinctions observed with KNN.

- This aligns with our expectation that these classes represent more general groupings rather than nuanced differences.
- K-Means, being an aggregate-focused algorithm, is naturally more sensitive to broader, larger-scale patterns, which explains why it captures general separations better than specific regional distinctions.

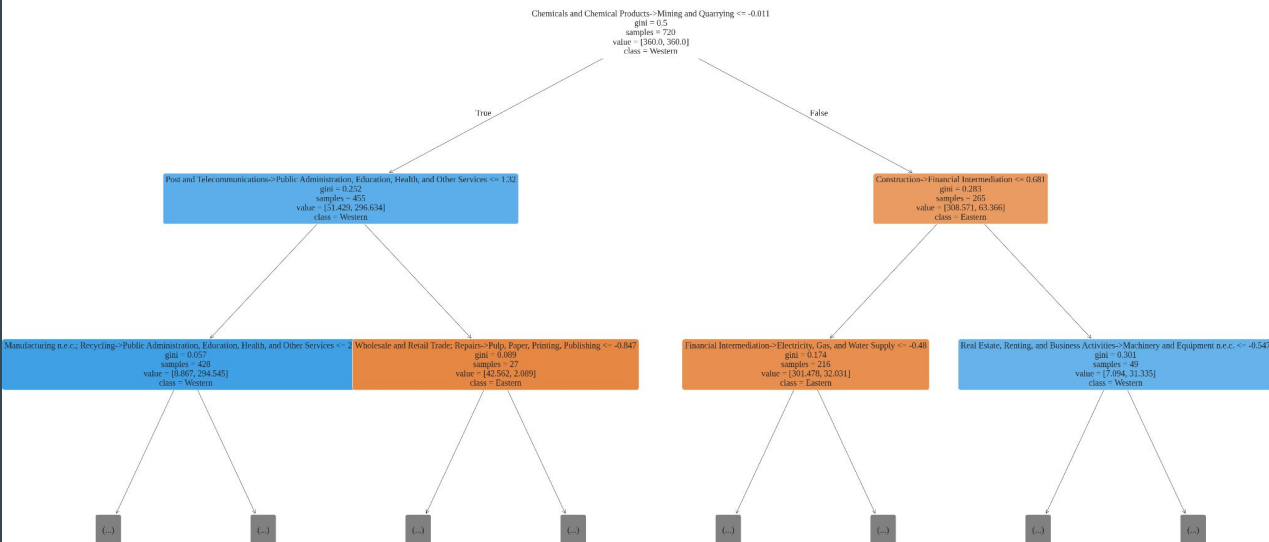


New Decision Trees

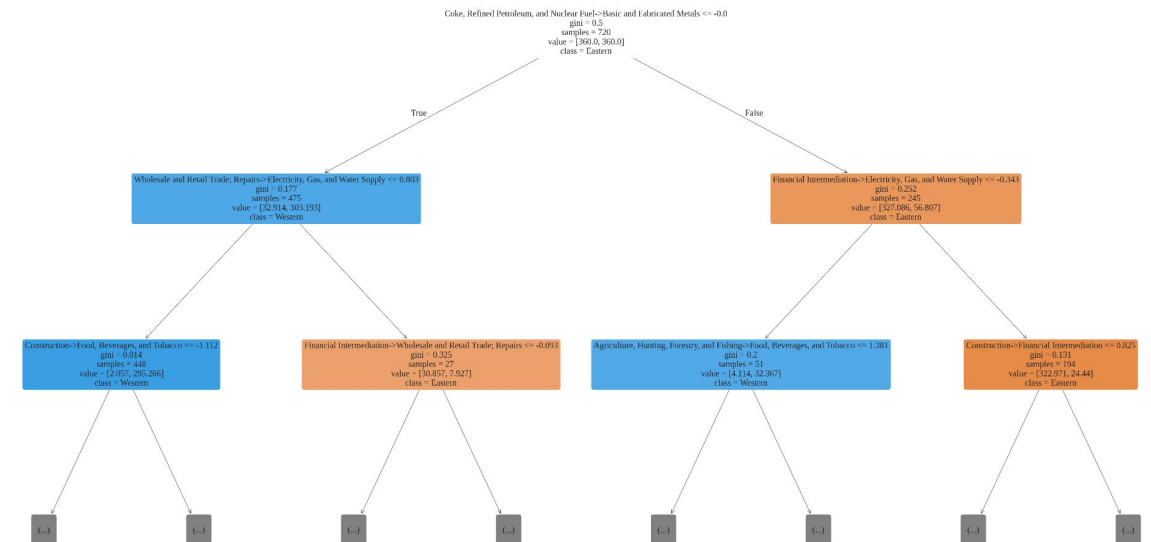
Political and Geographic Trees are structured **extremely differently**.

- Top splits are completely different
- Feature Importance also takes on a whole different structure.
 - Full Visualizations are available in the notebook.

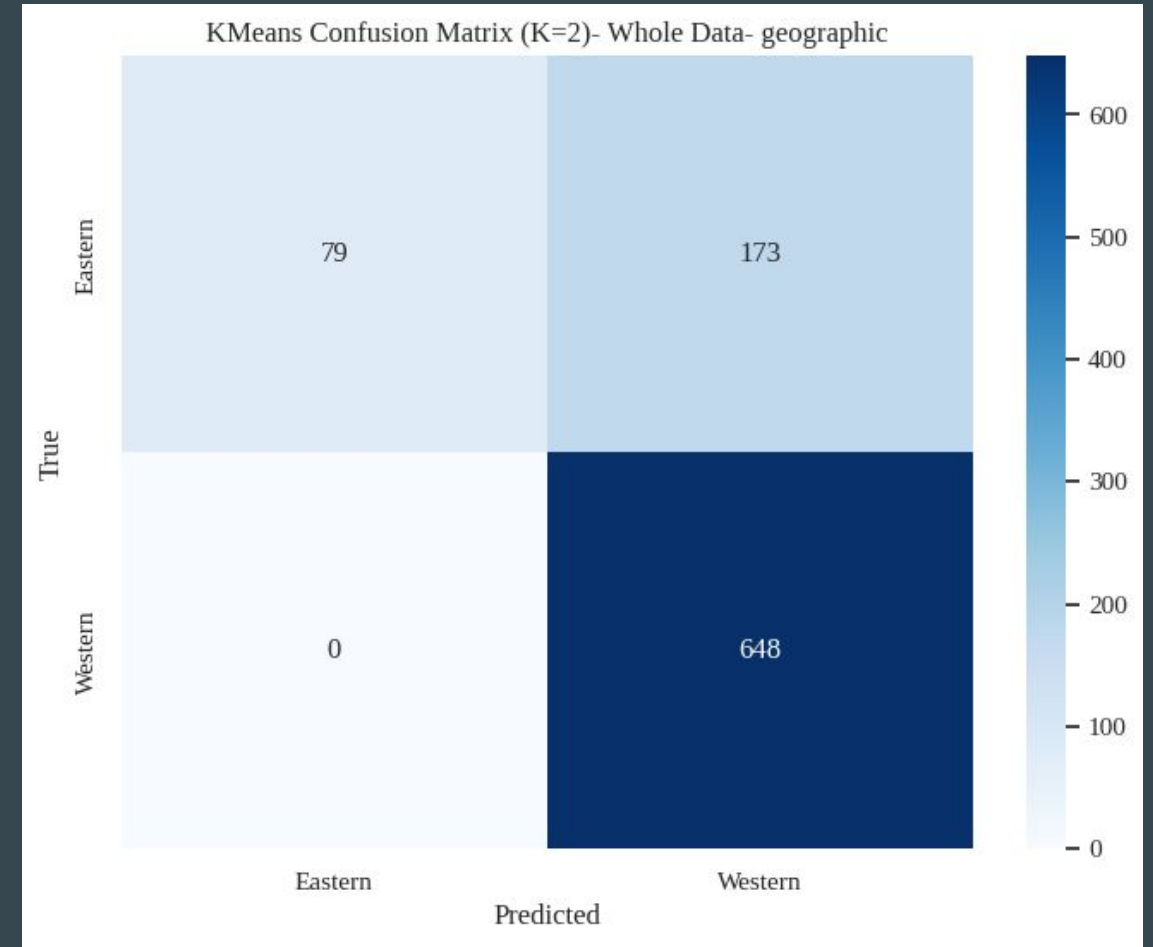
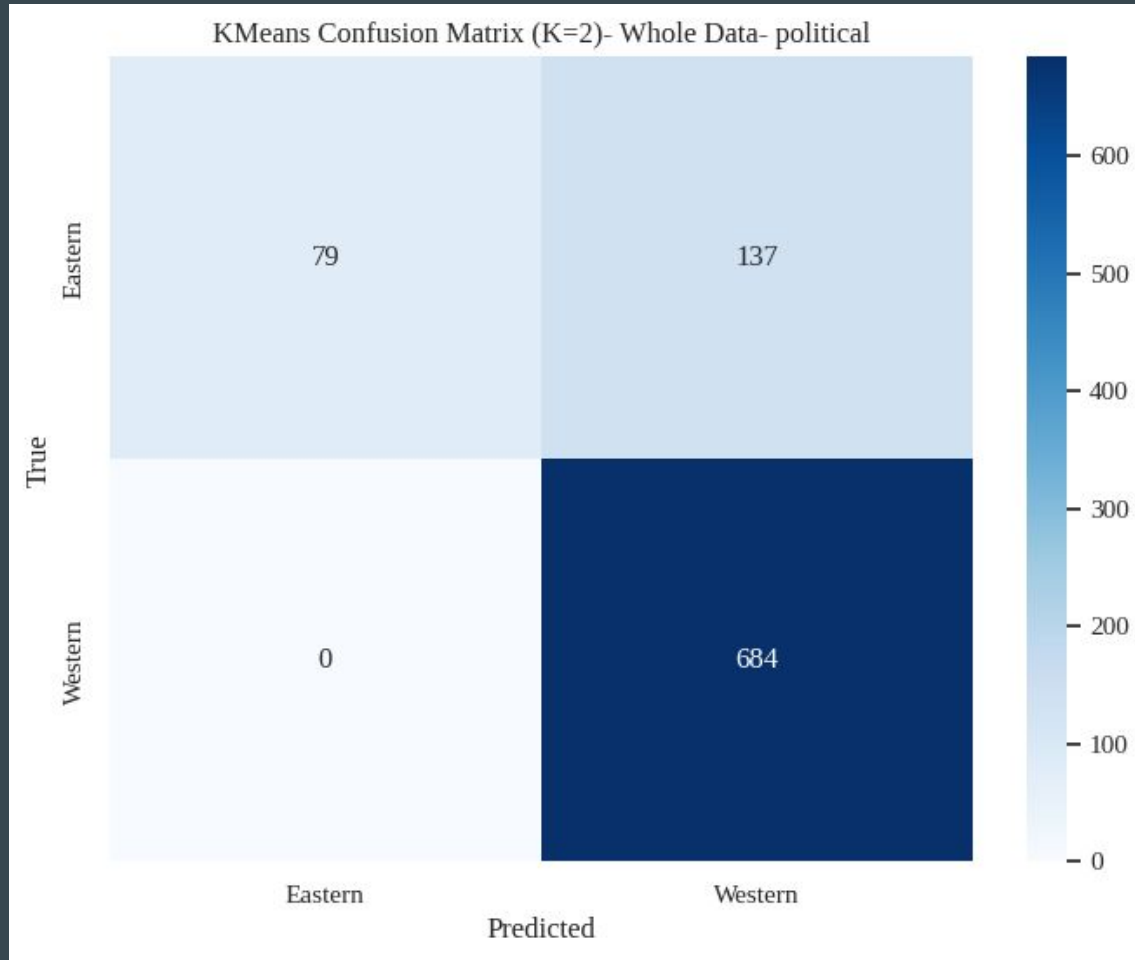
Decision Tree
Depth = 5, leaves = 15 - geographic



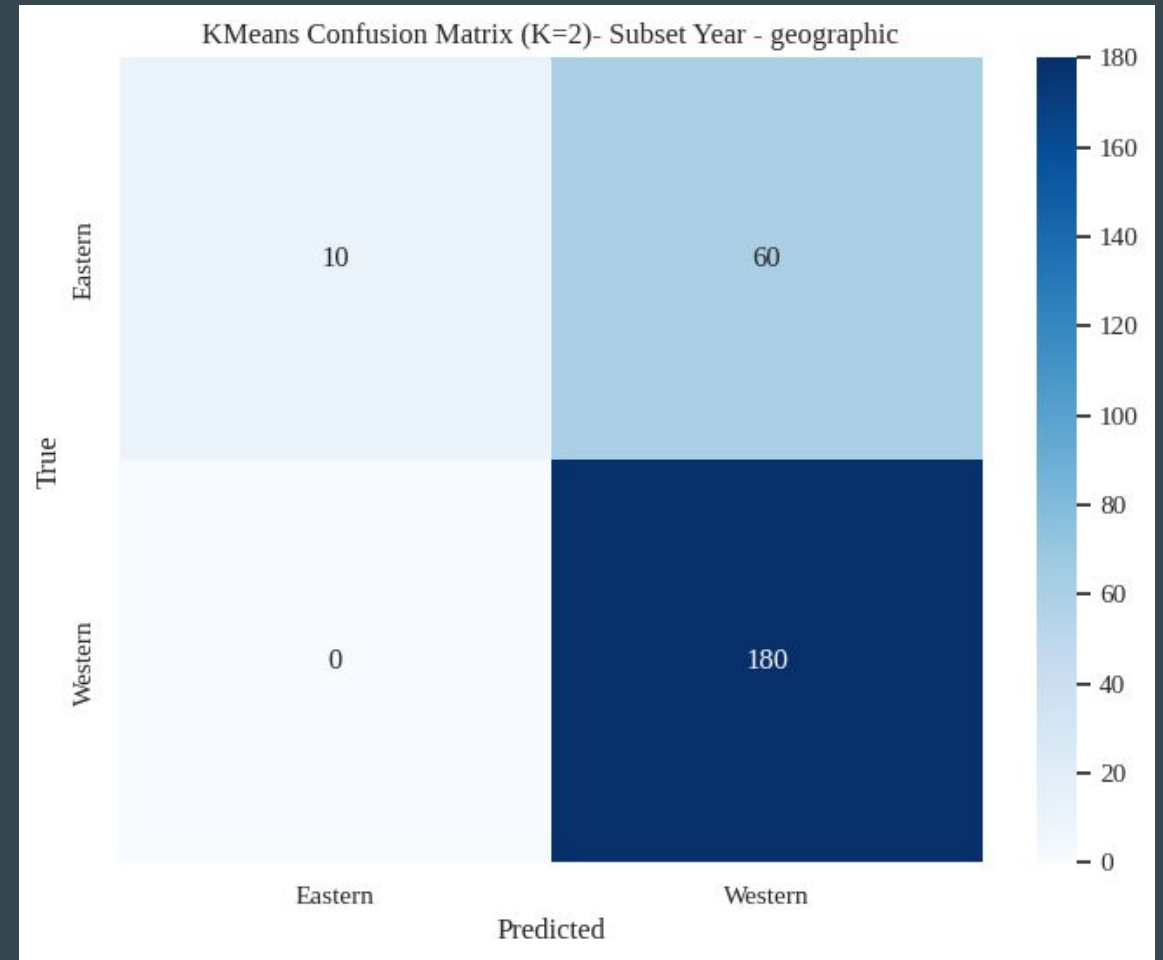
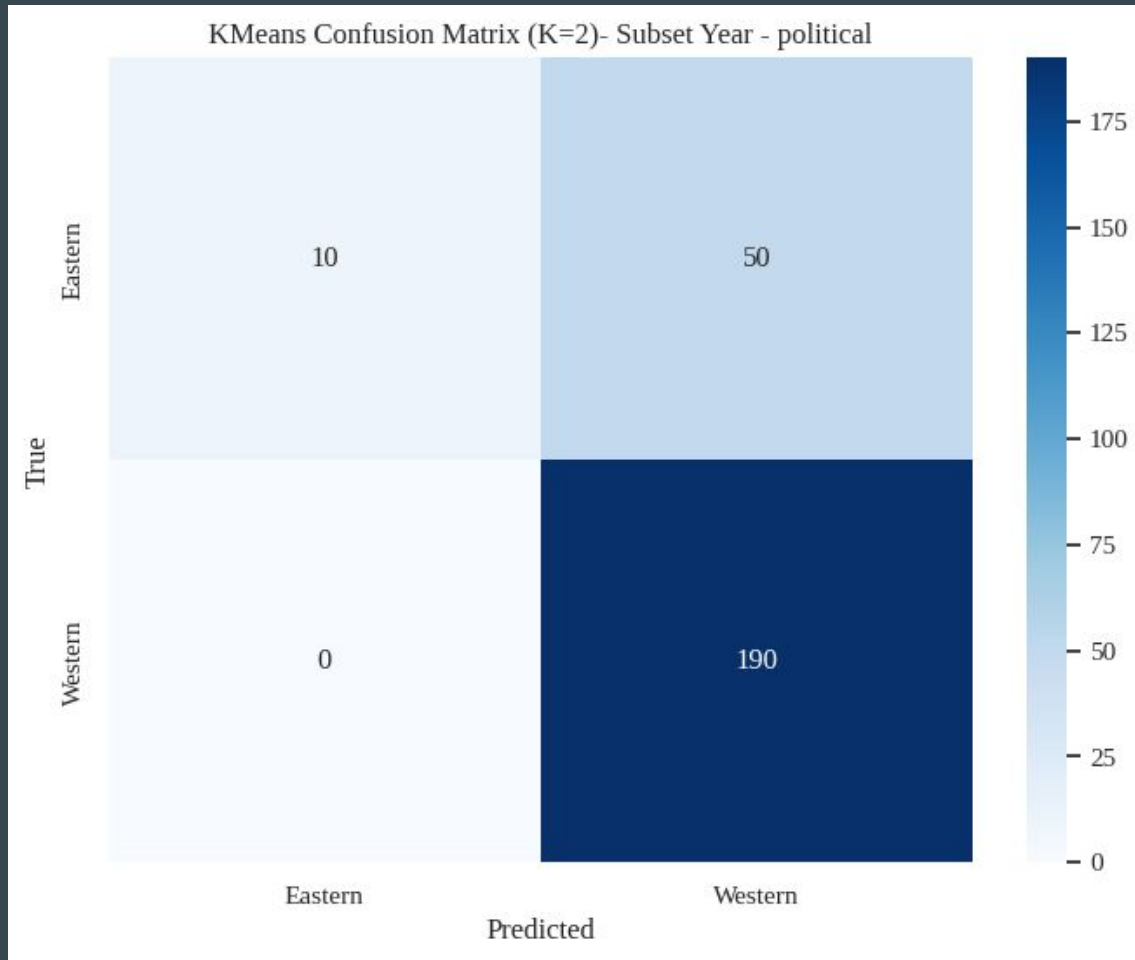
Decision Tree
Depth = 3, leaves = 8 - political



Clustering Results (2 classes) Whole Data



Clustering Results (2 classes) Forecasting



Clustering Observations

- Political Labels

- All Data

- === Confusion Matrix Metrics (Whole Data) ===
- Accuracy: 0.8477777777777777
- Precision (macro): 0.9165651644336175
- Recall (macro): 0.6828703703703703
- F1 Score (macro): 0.7222816600033786
- -----

- Forecast Data

- === Confusion Matrix Metrics (Subset Year) ===
- Accuracy: 0.8
- Precision (macro): 0.8958333333333333
- Recall (macro): 0.5833333333333334
- F1 Score (macro): 0.584717607973422
- -----

- Geographic Labels

- All Data

- === Confusion Matrix Metrics (Whole Data) ===
- Accuracy: 0.8077777777777778
- Precision (macro): 0.8946406820950061
- Recall (macro): 0.6567460317460317
- F1 Score (macro): 0.679787100582224
- -----

- Forecast Data

- === Confusion Matrix Metrics (Subset Year) ===
- Accuracy: 0.76
- Precision (macro): 0.875
- Recall (macro): 0.5714285714285714
- F1 Score (macro): 0.5535714285714286
- -----

Clustering Analysis

- K-Means favors large, structurally similar clusters, resulting in a dominant Western cluster.
- Average accuracy, probably due to the class imbalance
 - The model tends to predict the Western class by default, only switching to Eastern when highly confident.
- Low recall for non-Western groups
 - Smaller or structurally distinct regions (Asia-Pacific, Eastern) are not well separated in raw I/O structure.
- Precision > Recall
 - When the model identifies Eastern economies, the predictions are usually accurate.
 - However, the model is not very sensitive to these groups and rarely assigns them.
- Geopolitical and geographic labels show similar behavior
 - With K-Means, economic structures cluster more cleanly into East vs. West rather than Europe–Asia–Americas, reflecting the algorithm's aggregate nature.

Conclusion

There are clear structural differences in economic systems reflected in the Leontief inverse matrices. KNN, being a more local classification method, is able to capture these nuanced regional distinctions and differentiate smaller regions more effectively.

In contrast, K-Means, which relies on broad aggregation, struggles when differences between regions are subtle; as a result, it tends to form large, dominant clusters, especially under class imbalance.

- Given these properties, K-Means is better suited to identifying general economic structural groupings, but not specific geopolitical or geographic regions.
 - This leads to the model overwhelmingly assigning observations to a Western cluster, highlighting the difficulty of clustering regions whose economic structures vary only in nuanced ways.

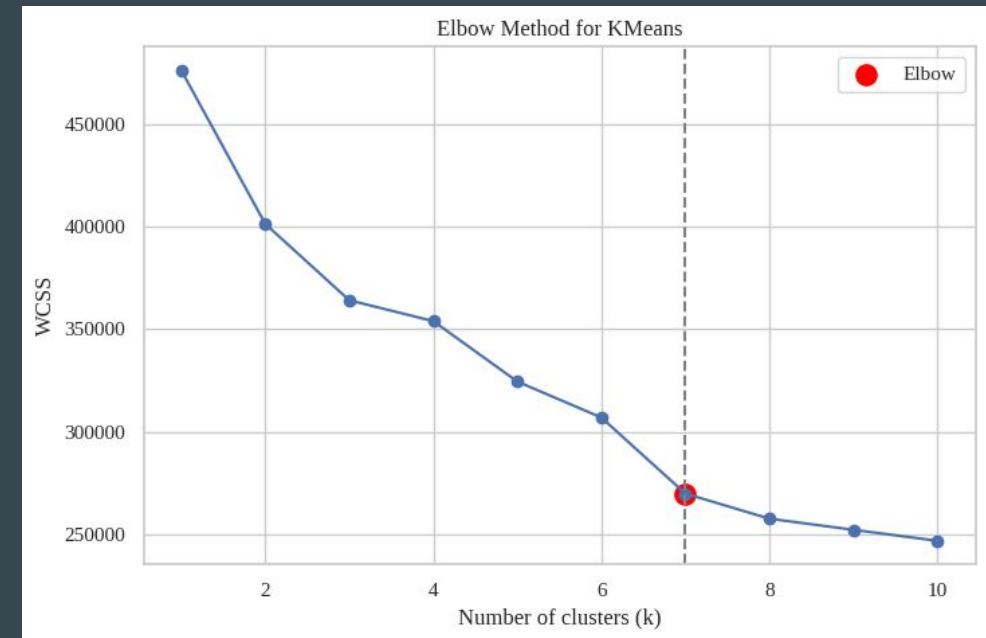
Future steps

Classification

- Obtain comparable time-series data for additional countries within each regional class.
- Evaluate the robustness of the trained models on unseen countries to test generalization.
 - Current dataset limitations (small sample size) prevented this analysis, but expanding the dataset would allow stronger validation.

Clustering

- Analysis of the WCSS metric suggests an optimal $K = 7$, indicating potential for more detailed structural groupings.
 - These clusters may reflect either general economic structure patterns or finer regional subdivisions beyond the East–West/Europe-Asia-America distinction.



References (more in speaker notes)

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